

Orientation solidification map through laser scan strategy Engineering for additively manufactured stainless steels

Foroozan Forooghi^{a,b,*}, Ayda Shahriari^a, Parisa Moazzen^a, Mohsen Keshavarzan^{a,b}, Nana Ofori-Opoku^b, Mohsen Mohammadi^a

^a Marine Additive Manufacturing Centre of Excellence (MAMCE), University of New Brunswick, Fredericton, NB E3B 5A1, Canada

^b Material Science & Engineering and Brockhouse Institute for Materials Research, McMaster University, Hamilton, ON L8S 4L7, Canada

ARTICLE INFO

Keywords:

Laser powder bed fusion (LPBF)
Scan Strategy
Three-Dimensional Finite Element Model
Orientation Solidification Map
316L Stainless Steel

ABSTRACT

Using a three-dimensional multiphase finite element model combined with a multiscale characterization framework, this study investigates the solidification characteristics and grain orientations of 316L stainless steel (316LSS) during laser powder bed fusion (LPBF) under different laser scanning strategies. Three common patterns—meander, stripe, and chessboard—were evaluated to capture the associated thermal profiles and melt pool (MP) dynamics. While conduction dominated across all strategies, the stripe and chessboard patterns showed greater susceptibility to keyhole and transition melting modes. Thermal simulations revealed variations in remelting depth: the stripe pattern exhibited the deepest remelting ($\sim 150\ \mu\text{m}$), while the meander pattern showed the shallowest ($\sim 50\ \mu\text{m}$), influencing grain refinement and morphology. Crystallographic orientation analysis indicated that interactions between keyhole, transition, and conductive MPs altered preferred growth directions. The meander strategy promoted [101] and [111] orientations, whereas the stripe and chessboard strategies showed stronger alignment toward [111]. By correlating thermal parameters—temperature gradient (G) and solidification rate (R)—with local crystallographic orientation and grain structure, a solidification orientation map was developed. This map demonstrates how scan strategy-driven MP interactions control microstructural features, providing a predictive tool for grain morphology and texture design in LPBF-316LSS components.

1. Introduction

The next industrial revolution is being driven by advancements in manufacturing techniques, particularly additive manufacturing (AM) which enables the production of parts from digital models through layer-by-layer deposition of material [1]. Among AM processes, laser powder bed fusion (LPBF) allows fabrication of complex geometries and customized components across multiple industries. In this technique, a high-power laser selectively melts fine powder particles layer by layer [2,3]. The high energy density in a small, localized area leads to rapid heating and cooling rates and multiple thermal cycles [4–6]. This produces a non-equilibrium solidification microstructure characterized by hierarchical features and diverse grain morphologies with different crystallographic orientations [7,8].

Solidification in LPBF has been extensively studied to investigate various aspects, such as crystal growth rates [9,10], grain orientation and texture control [10–14], and the development of process-property

maps for different materials [15]. Microstructural evolution is closely linked to quantitative process parameters such as laser power, scan speed, hatch spacing, and layer thickness, which alter melt pool (MP) morphology by modifying the energy input to the powder [16,17]. In addition, scan strategy, as a qualitative process parameter, governs the laser path during layer scanning and influences the volumetric energy input, thermal distribution, and degree of remelting between scan vectors [18,19]. Prior studies [20,21] have shown that scan strategy affects dendrite orientation, grain morphology, and columnar–equiaxed transitions, while also influencing temperature distribution, thermal stresses, distortion, and defect formation [22–25]. Dinda et al. [26] reported that unidirectional laser scanning in Ni-based alloys promotes dendrites oriented 60° to the substrate, whereas bidirectional scanning rotates dendrite orientation by 90° in each successive layer. Kirka et al. [6] observed a transition from columnar to equiaxed morphology in a Ni-based superalloy when the scan strategy changed from raster to spot, which increased energy density and reduced thermal gradients.

Experimental monitoring of thermal fields, such as in-situ thermal

* Corresponding author.

E-mail address: F.Forooghi@unb.ca (F. Forooghi).

<https://doi.org/10.1016/j.matdes.2025.114880>

Received 14 July 2025; Received in revised form 24 September 2025; Accepted 3 October 2025

Available online 5 October 2025

0264-1275/© 2025 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

Nomenclature

BD	Building Direction
L	Layer thickness (μm)
D	Melt Pool Depth (μm)
H	Hatch Distance (μm)
W	Melt Pool Width (μm)
T	Temperature (K)
k	Thermal Conductivity ($\text{W/m}\cdot\text{K}$)
Q	Heat source per unit volume (W/m^3)
ρ	Density (kg/m^3)
c_p	Specific heat ($\text{J/kg}\cdot\text{K}$)
t	Time (s)
σ	Stefan-Boltzmann ($\text{W/m}^2\cdot\text{K}^4$)
ϵ	emissivity
h	convection ($\text{W}/(\text{m}^2\cdot\text{K})$)
T_∞	Chamber Temperature (K)
T_L	Liquidus Temperature (K)
$T_{\max}$	Maximum Temperature during Melting (K)
$G \times R$	Cooling Rate (K/s)
G	Temperature Gradient (K/m)
R	Solidification Rate

imaging, is limited by insufficient spatial and temporal resolution, making it difficult to capture rapid cooling rates and steep temperature gradients during LPBF [27–29]. Consequently, numerical simulations have become indispensable, complementing experimental efforts by offering detailed insights into thermal history and microstructure evolution. The finite element (FE) method has been widely used for this purpose [30–33]. Moazzen et al. [34] developed a numerical thermal model to examine the influence of process parameters and scan strategies on microstructural features of LPBF-316LSS. Their findings showed that meander (continuous) scan strategy with constant energy input produced high-density components, with the finest microstructure in comparison with other scan strategies. Parry et al. [35] used a thermo-mechanical model to demonstrate that unidirectional strategies reduce residual stresses compared to alternating ones due to differences in thermal history. Azizi et al. [36] examined the influence of building direction and thermal gradients in the microstructure evolution of dilute Al-Si alloy processed by LPBF by coupling a FE model to a phase-field simulation.

On the one hand, understanding the role of process parameters in LPBF is essential for controlling solidification behavior and achieving tailored mechanical properties. Among these parameters, the scanning strategy plays a particularly critical role in dictating MP dynamics and, consequently, grain morphology, size, and crystallographic orientation in austenitic stainless steels such as 316L. However, the mechanisms by which different scanning strategies affect MP evolution and cause deviations from the expected crystallographic orientations remain poorly understood. On the other hand, capturing the thermal history required to study these mechanisms can be achieved more effectively through numerical thermal modeling, which provides detailed insights into melt pool behavior and microstructure evolution.

These limitations highlight the need for an integrated approach combining numerical modeling and experimental validation. To address this gap, we developed an orientation solidification map using advanced numerical modeling and multiscale characterization. A three-dimensional, multi-layer FE model was employed to simulate thermal evolution in LPBF, providing insights into MP geometry, temperature distribution, thermal gradients, and cooling rates for three scanning strategies: meander, stripe, and chessboard. Complementary electron backscatter diffraction (EBSD) characterization was used to correlate thermal history with grain structure. The combined approach revealed

how scanning strategies influence deviations in crystallographic orientation and grain morphology in LPBF-316LSS, supporting the development of an orientation–solidification map.

2. Experimental Methodology

2.1. LPBF fabrication strategies

In this study, a powder bed was formed using virgin gas-atomized 316LSS powder (supplied by GE), with a particle size distribution ranging from 15 to 65 μm and an average size of $21.84 \pm 0.76 \mu\text{m}$, as shown in Fig. 1a. The chemical composition of 316LSS was 0.02 wt% C, 13 wt% Ni, 16.8 wt% Cr, 0.85 wt% Si, 0.2 wt% O, 2.2 wt% Mo, 0.04 wt% N and balanced Fe. The additive manufacturing process was conducted using a GE Concept Laser M2 Cusing LPBF machine. The machine was equipped with a 400 W CW modulated ytterbium fiber laser (1070 nm wavelength and effective spot size of 72 μm) along with CL WRX CONTROL software and CL WRX “Parameter Editor”. To maintain the oxygen level below 200 ppm during the process and prevent oxidation of the molten material and powder, the chamber was flushed with nitrogen gas.

The LPBF-316LSS cubic specimens, with dimensions of 10 mm $\times$ 10 mm $\times$ 10 mm, were additively manufactured on a substrate composed of the same material, using the process parameters specified in Table 1. These parameters were adopted from the comprehensive design of experiments in reference [34], where twenty parameters with different energy levels—44.4, 47.6, 50.0, and 60.9 J/mm^3 —were evaluated across five distinct scan strategies. Parameters with a volume energy density of 60.9 J/mm^3 , resulting in higher density (lower porosity), were selected as the optimized parameters for this work along with three common scan strategies: meander, stripe, and chessboard, as shown schematically in Fig. 1b. The scanning vectors in each layer were rotated by 67°, with the same rotation applied to successive layers. The use of rotational strategies introduces complexity into the thermal fields due to the varying scanning and hatching directions in each layer, influencing microstructure and texture formation [22].

In the meander scan strategy, the laser follows a continuous back-and-forth path across the scan area, maintaining a consistent scan vector length along parallel lines. At the edges, where the laser transitions between perpendicular sides, the scan path is slightly shortened. In the stripe scan strategy, the scan area is partitioned into multiple stripes, each scanned individually in a linear fashion. Although this leads to shorter scan vectors than in the meander pattern, the presence of repositioning jumps between stripes increases the scan time. The chessboard scan strategy divides the area into small blocks. The laser scans each block alternately in a checkerboard-like pattern, with scan vectors rotating by 90° between layers. Additionally, the entire scan pattern is rotated by 67° for each new layer to promote grain refinement. While scan vector lengths vary slightly across blocks, the frequent jumping between regions extends the overall scan time.

2.2. Microstructural characterization

Microstructural analysis was conducted on specimens sectioned from the center of the cuboidal samples in the Y-Z (build direction) plane (Fig. 1b) using an optical microscope (MP) and scanning electron microscope (SEM). Metallographic preparation involved grinding with silicon carbide papers of progressively finer grit sizes down to 4000 grit, followed by polishing with colloidal silica suspensions to achieve a 1 μm finish, followed by final polishing with colloidal silica (0.05 μm). Optical and SEM characterizations were carried out after etching with Beraha's tint etchant, consisting of 100 mL of water, 20 mL of HCl, and 0.5–1.0 g potassium metabisulfite.

SEM characterization of the powder and specimens at both low and high magnifications was conducted using an FEI Scios 2 HiVac dual-beam SEM equipped with an Oxford EBSD Symmetry® S3 detector

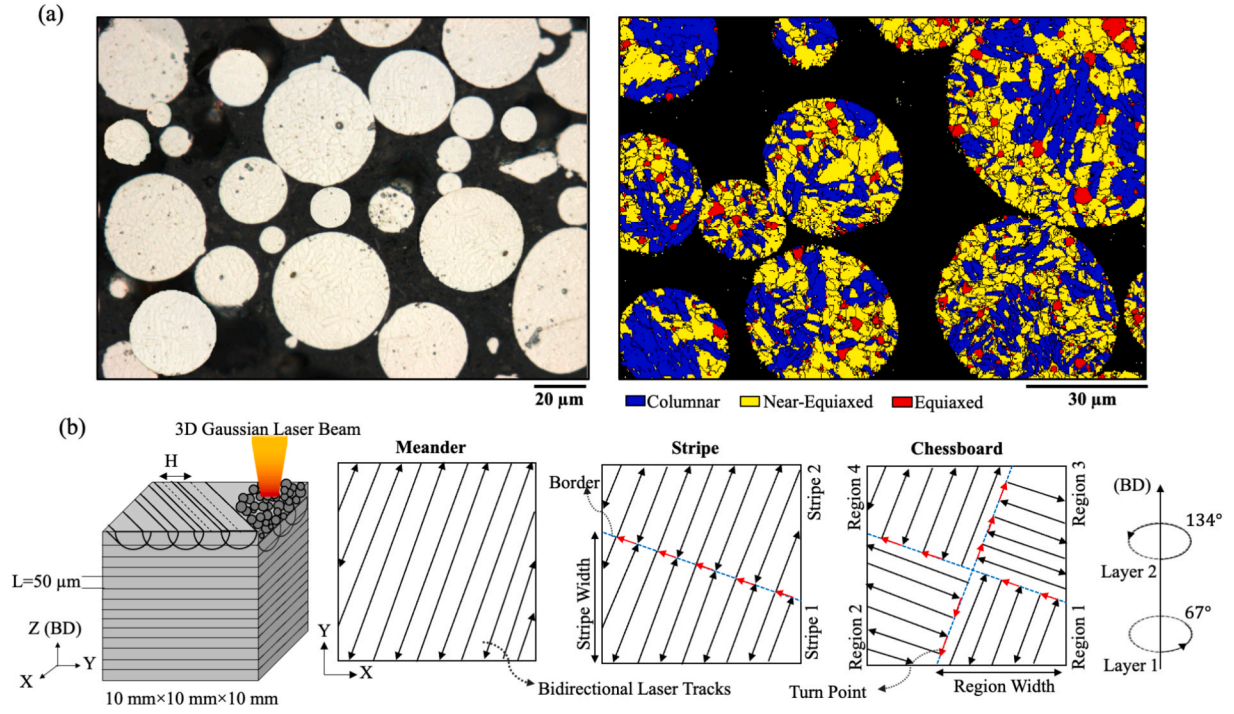


Fig. 1. (a) Optical microscope image of virgin 316LSS powder with EBSD map showing grain aspect ratio; (b) schematic of the sample and scan strategies (meander, stripe, chessboard).

Table 1
Processing parameters for LPBF-316LSS specimens used in this study.

Parameter	Symbol	Value
Laser Power	P	370 W
Scan Speed	V	1350 mm/s
Hatch Space	H	90 μm
Layer Thickness	L	50 μm
Laser Diameter	D	72 μm
Stripe/Chessboard Width	—	10 mm

(CMOS sensor, 1244×1024 pixels). The system was integrated with Oxford AZtecHKL software. Measurements were carried out over areas of $\sim 1 \text{ mm}^2$ with a step size of 0.2 μm at a tilt angle of 70° . The resulting diffraction patterns were further analyzed using TSL® OIM Data Analysis software.

3. Theory and computational framework

To study the transient thermal behavior in the LPBF process, a three-dimensional, multiphase thermal model was developed using FE method at the macroscale. The model consisted of the formulation of heat transfer equations and an effective heat source model for energy deposition, followed by an explanation of the numerical implementation.

3.1. Governing equations

In laser-based additive manufacturing methods such as LPBF, heat transfer is dominated by conduction, with additional contributions from radiation, convection, and evaporation losses. However, it is noteworthy that conduction plays a significant role in overall heat transport. The governing equation for three-dimensional heat transfer is described by the classical heat equation, which gives temperature distribution in x, y, z as presented in Eq. (1) [37–42]:

$$\frac{\partial}{\partial x} \left(k_x(T) \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k_y(T) \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(k_z(T) \frac{\partial T}{\partial z} \right) + Q(x, y, z, t) = \rho c_p \frac{\partial T}{\partial t} \quad (1)$$

In this equation k_x, k_y, k_z represent the thermal conductivity in different directions, T denotes the current temperature at any location within the model, Q represents the internal heat generation per unit volume measured in W/m^3 , ρ signifies the density, and c_p is the specific heat of the material. The variable t is time (s), corresponding to the duration of laser–powder interaction. Heat transfer due to fluid motion within the melt pool, referred to as thermocapillary (Marangoni) convection, can be effectively captured by employing anisotropic thermal conductivity. This is implemented by introducing a temperature-dependent enhancement factor, α_i , which scales the thermal conductivity k along different directions. Calibration of α_i is carried out by comparing simulated melt pool dimensions with experimental cross-sections until convergence is achieved, ensuring accurate prediction of MP geometry [43]. Using anisotropic conductivity improves the accuracy of MP dimension predictions, as described in Eq. (2) by [44,45]:

$$k_i(T) = \alpha_i k, \quad i = x, y, z \quad (2)$$

where α_i represents the enhancement factors of k , is calibrated by minimizing the deviation between the estimated and experimental MP size.

$$\alpha_i = \begin{cases} 1 & T < T_L \\ \alpha_x = \alpha_z = 10, \quad \alpha_z = 20 & T > T_L \end{cases} \quad (3)$$

The accuracy of the heat transfer model is primarily influenced by the precise estimation of heat input and the selection of the heat source model. The moving laser scanning, serving as the heat source in this work, is modeled using a volumetric Gaussian heat source [46]. This model is extensively employed for its effectiveness in capturing the energy distribution within a laser beam. It features an exponential decay with depth (z -direction) and is expressed as:

$$q(x, y, z, t) = \frac{2\eta P}{\pi R_0^2 S} \exp\left[-2\frac{r^2}{R_0^2}\right] \exp\left[-\frac{|z|}{S}\right] \quad (4)$$

where r represents the radial distance from beam center. R_0 denotes the spot radius, η is the laser absorptivity of the material, $|z|$ is the absolute value of the z -coordinate, and S is the optical penetration depth of the laser. Considering the losses, heat supplied, and temperature distribution, the thermal equilibrium equation of the LPBF process can be stated as [25,47]:

$$-k\left(\frac{\partial T}{\partial n}\right) = \sigma\epsilon(T^4 - T_\infty^4) + h(T - T_\infty) \quad (5)$$

In the heat loss equation –Eq. (5)–, n is the normal vector of the surface, σ denotes the Stefan-Boltzmann constant for radiation, ϵ represents the emissivity, h signifies convection, and T_∞ stands for the chamber temperature. The bottom surface is considered adiabatic.

3.2. Solid-liquid phase change

The total enthalpy (H) is implemented in this study to consider phase change during melting and solidification. This approach involves compensating for the latent heat of fusion by augmenting the specific heat between the solidus and liquidus. This eliminates the need to explicitly track the solid-liquid interface [48]. This adjustment ensures that the additional heat absorbed by the material in this temperature range equals the latent heat of fusion. Consequently, within these temperature intervals, there is a significant and a sharp increase in effective specific heat [45].

$$H = \begin{cases} \int c_p dT, & T < T_s \\ \int c_p dT + \frac{T - T_s}{T_l - T_s} L, & T_s < T < T_l \\ \int c_p dT + L, & T \geq T_l \end{cases} \quad (6)$$

where L is the latent heat of fusion, c_p is the specific heat capacity, T_l and T_s are the liquidus and solidus temperatures, respectively.

3.3. Numerical implementation

A numerical model was created and simulated in Abaqus, utilizing the FE method to accurately replicate the LPBF process. To simplify the modelling process, the entire powder bed was treated as a homogeneous and continuous medium, to simplify the mathematical formulation. It was assumed that layer contraction was negligible, maintaining a constant layer thickness before and after laser scanning. The thermophysical properties of LPBF-316LSS—thermal conductivity, density, and specific heat capacity—were modeled as temperature-dependent values [49–54], accounting for variations in powder, liquid, and solid states near their respective melting and solidification temperatures.

The three-dimensional computational domain consisted of a powder bed deposited on top of a solid 316LSS substrate (Fig. 2a). The powder bed measured $2700 \times 2700 \times 250 \mu\text{m}^3$, corresponding to five layers of $50 \mu\text{m}$ thickness each. A solid substrate of $2700 \times 2700 \times 900 \mu\text{m}^3$ was placed beneath the powder. Powder layers were incrementally added in a stepwise manner to replicate the layer-by-layer deposition of the LPBF process.

To ensure accuracy while maintaining computational efficiency, the region directly affected by the laser was discretized using high-resolution DC3D8 heat transfer brick elements with element sizes of

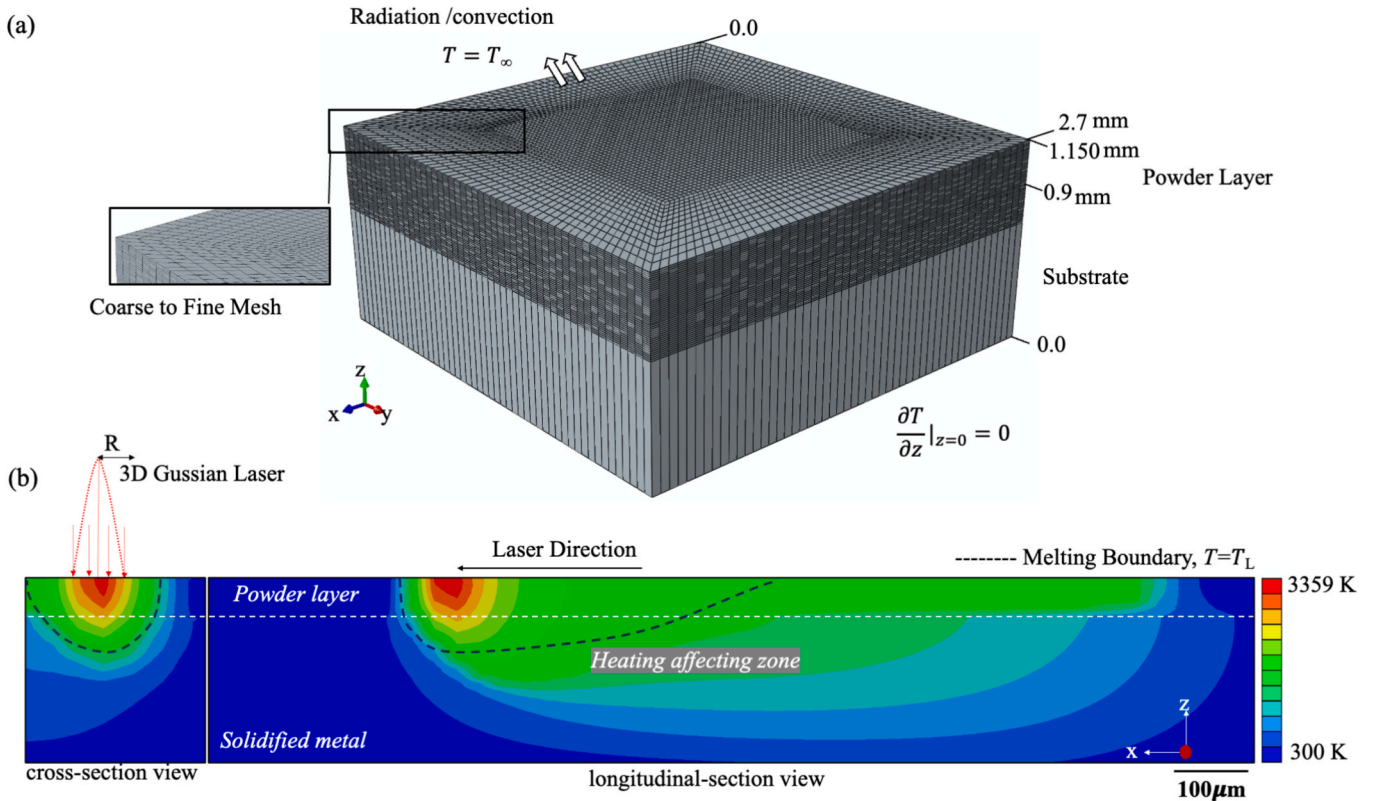


Fig. 2. (a) Computational domain and boundary conditions of the FE model; (b) cross-sectional and longitudinal temperature distributions showing the MP geometry.

$30 \times 30 \times 10 \text{ } \mu\text{m}^3$, using a time step of $\Delta t \approx 10^{-6}$ s. The scanning surface ($1500 \times 1500 \text{ } \mu\text{m}^2$) was finely meshed to capture steep thermal gradients, while coarser elements were used in surrounding regions to reduce computational demand.

Adiabatic boundary conditions were applied to the bottom surface ($\partial T / \partial z|_{z=0} = 0$), while convective heat loss was imposed on the top surface by imposing a constant ambient temperature $T = T_\infty$. A moving Gaussian heat source was employed to simulate the laser's scanning path.

Fig. 2b shows the cross-sectional and longitudinal temperature distributions beneath the laser track. The MP boundary was defined by isothermal contours at the liquidus temperature set to 1723 K for LPBF-316LSS. As the laser moves, heat is conducted away from the MP into the surrounding powder particles, the solidified material below, and the deeper substrate, while the liquid MP subsequently solidified. The temperature gradients and isothermal contours highlight this multidirectional heat diffusion, resulting in a semi-elliptical MP shape. This thermal simulation provides insight into MP behavior through predicting of temperature distribution, melt pool geometry, solidification rates (R), temperature gradients (G). These parameters are directly linked to solidification conditions (G/R) and microstructure evolution, thereby enabling assessment of the impact of scanning strategies on grain morphology and orientation, even in the absence of detailed flow simulations [25].

Laser heat transfer and scanning strategies were simulated using the DFLUX subroutine to control energy flux input. Phase transformations between powder, liquid, and solid states were modeled using the USFLD subroutine, ensuring accurate representation of thermal and phase transition during LPBF. A recoating time of 12 s—defined as the delay between powder deposition and scanning of the previous layer—was introduced to allow full cooling of each layer before the next scan.

Temperature-dependent thermophysical properties of LPBF-316LSS, including density, thermal conductivity, and specific heat capacity, were obtained from JMatPro software to ensure accurate representation of dynamic thermal conditions and phase transformations (Table 2, Fig. 3). These properties play a critical role in predicting thermal histories and melt pool evolution during the LPBF process.

4. Results and discussions

4.1. Thermal profile and geometry of the MPs

Before simulating MP geometries for each scan strategy, the experimental measurements were obtained from OM images. These images were taken from the central region of each scan strategy on the Y–Z

Table 2
Thermophysical parameters and other variables used in the FE thermal model [49–54].

Physical variables	Symbol	Value and unit
Liquidus Temperature	T_L	1723 K
Solidus Temperature	T_s	1658 K
Chamber temperature	T_∞	298 K
Emissivity	ε	0.4
Convection coefficient	h	$700 \text{ W}\cdot\text{m}^{-2}\cdot\text{K}^{-1}$
Latent heat of fusion	L_f	$2.6 \times 10^5 \text{ J}\cdot\text{kg}^{-1}$
Laser absorptivity	η	0.458
Penetration depth	S	$50 \text{ } \mu\text{m}$
Specific heat	c_p	Temperature dependent (Fig. 3)
Powder porosity	ϕ	0.4
Density of solid	ρ_{solid}	Temperature dependent (Fig. 3)
Density of powder	ρ_{powder}	$\rho_{\text{solid}} \times (1 - \phi)$
Thermal conductivity of solid	k_{solid}	Temperature dependent (Fig. 3)
Thermal conductivity of powder	k_{powder}	$k_{\text{solid}}(1 - \phi)$
		$\left(1 + (0.02 \times 10^{2(\phi-0.3)})\right) \frac{k_{\text{solid}}}{k_{\text{gas}}}$
Thermal conductivity of argon gas	k_{gas}	0.016 W/m

planes of cuboidal samples, each measuring $10 \text{ mm} \times 10 \text{ mm}$. The OM images were then processed using ImageJ to quantify the MP dimensions. For each sample, ~ 200 measurements were taken across an $800 \times 800 \text{ } \mu\text{m}$ area at the core and at multiple heights to assess consistency and geometry. MP depth was determined by first locating the deepest vertical point, then drawing a perpendicular line from that point to a horizontal reference line to define the lateral extent of the MP. procedure is schematically illustrated in Fig. 4a along with the statistical values for both MP width and depth.

To calibrate the thermal model, a preliminary single-layer simulation was performed for the meander scan strategy and compared with experimental cross-sectional SEM measurements under identical conditions [48] (Fig. 4b). Simulated MP geometries were extracted from the center of the scan track, while experimental dimensions were quantified from SEM images. For simulations, values were averaged across multiple cross-sections to capture spatial variations. The resulting MP width and depth showed close agreement with experiment (Table 3), confirming the model's fidelity. The same procedure was then applied to stripe and chessboard strategies (Fig. 4c–d), where simulated MP profiles were likewise consistent with experimental cross-sections.

This single calibration using region-specific thermal fields for the meander strategy, followed by validation against stripe and chessboard experiments, demonstrates the robustness and predictive capability of the developed model. The consistent agreement in MP shape and boundary definition across all strategies confirms its reliability for analyzing LPBF melt pool behavior and guiding microstructural interpretation.

Following the geometric validation of MP profiles (Fig. 4b–d), detailed thermal analysis were performed to evaluate temperature history, cooling behavior, and gradients during full single-layer scanning. The thermal simulations captured the evolution of temperature across the entire layer, with the monitoring point P strategically placed near the end of the scan in the meander and stripe strategies, and within the final region (R4) in the chessboard scan. This positioning reflects the zone of highest accumulated thermal exposure and is critical for evaluating temperature gradients, cooling rates, and solidification behavior at the end of each scanning cycle.

As illustrated in Fig. 5.a–c, the plots of temperature versus time at point P reveal the transient thermal behavior as the laser traverses the layer at the bottom of the MP. A sharp rise in temperature is observed during laser exposure, followed by a rapid decline after the beam passes. The cooling rate was obtained from the slope of the temperature–time curve between the peak and liquidus temperatures (1723 K for LPBF-316LSS), and is herein reported as $G \times R$. The simulations show that after the temperature drops below the liquidus point, the material enters a mushy zone, characterized by the coexistence of solid and liquid phases, before fully solidifying. The numerical cooling rates and temperature gradients at the top and bottom of the MPs in conductive mode at point P for the three different scanning strategies are also calculated and presented in Table 4.

Across all strategies, the MP bottom experienced higher cooling rates than the top due to efficient conduction into the solidified substrate, whereas the top primarily dissipated heat through slower radiation and convection. This strong gradient at the bottom favors columnar grain growth. As expected, the meander strategy produced the steepest gradients, followed by chessboard and stripe. The temperature gradient (G) was computed using the spatial derivatives of the temperature field –Eq. (7)– [55], where:

$$G = \sqrt{G_x^2 + G_y^2 + G_z^2}, \quad G_i = \frac{\partial T_i}{\partial t}, \quad i = x, y, z \quad (7)$$

Table 4 reveals that the temperature gradient follows a similar trend to the cooling rate, with the meander strategy producing the steepest gradients, followed by chessboard and stripe. However, the magnitude of the temperature gradient was consistently higher than the cooling

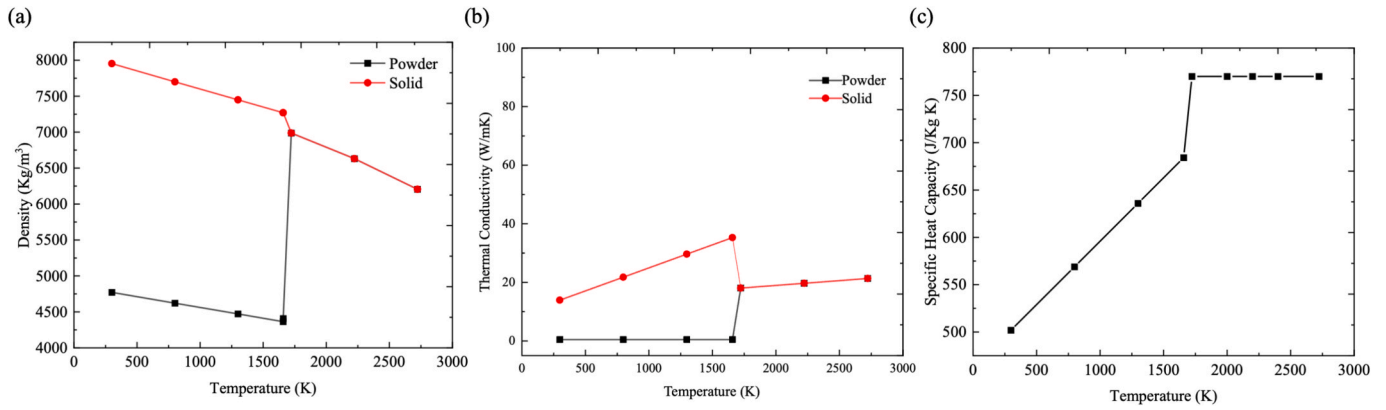


Fig. 3. Temperature-dependent material properties of 316LSS: (a) density, (b) thermal conductivity, and (c) specific heat capacity for powder and solid.

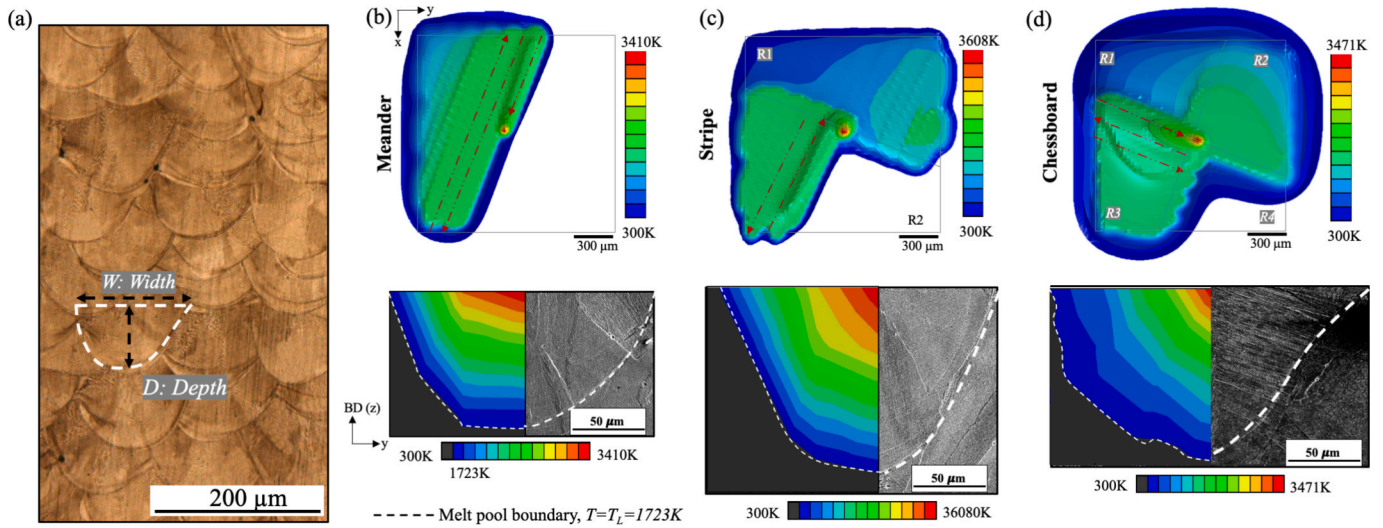


Fig. 4. (a) MP measurement workflow on OM image; (b–d) calibration/validation: simulation vs. SEM for meander, stripe, chessboard (dashed = experimental boundary).

Table 3

Comparison of simulated and experimental MP dimensions (conduction mode).

Strategy	Dimension	Experiment (μm)	Simulation (μm)	Error (%)
Meander	Width	135 ± 15	138 ± 12	2.22
	Depth	85 ± 20	90 ± 5	5.08
Stripe	Width	195 ± 50	199 ± 7	2.05
	Depth	140 ± 40	146 ± 4	4.28
Chessboard	Width	175 ± 40	178 ± 8	1.71
	Depth	120 ± 30	124 ± 10	3.33

rate, reflecting the strong spatial variation in temperature, particularly near the MP boundary.

The simulation results were further evaluated against OM observations of MPs formed under three different scanning strategies, as shown in Fig. 6a–c. These low-magnification cross-sectional images, oriented parallel to the build direction, reveal distinct MP geometries. In the meander strategy (Fig. 6a), MPs display regular semi-cylindrical morphologies with consistent overlap between adjacent tracks. In contrast, the stripe strategy (Fig. 6b) produces wider and deeper MPs, often showing over-melting and keyhole features, with irregular shapes and deeper substrate penetration caused by localized heat accumulation. The chessboard strategy (Fig. 6c) yields MPs shallower and narrower than stripe but larger than meander, with transition-mode features that combine conduction- and keyhole-like geometries—wider at the top and

narrower at the bottom [56].

These observations are consistent with the simulated cooling rates and thermal gradients (Table 4) and the simulated MP morphologies (Fig. 4). The meander strategy exhibited the highest cooling rates— 3.2×10^6 K/s (top) and 4.85×10^6 K/s (bottom)—linked to its continuous, efficient laser path and minimal idle time between scan vectors, which promote rapid heat dissipation, smaller MPs, and faster solidification (Fig. 4b). In contrast, the stripe and chessboard strategies produced slower cooling rates due to discontinuous scanning and frequent overlaps, which increased local heat accumulation and delayed solidification. For stripe, scanning of two sequential regions (R1 and R2) resulted in cooling rates of 1.1×10^6 K/s (top) and 1.93×10^6 K/s (bottom). For chessboard, four smaller regions (R1–R4) produced intermediate rates of 1.6×10^6 K/s and 2.8×10^6 K/s, respectively. These lower cooling rates correlate with larger MPs and longer solidification times, both of which can influence grain size in the solidified microstructure.

Fig. 6d further classifies MPs by depth-to-width ratio (D/W) into lack of fusion (LOF), conduction, transition, and keyhole modes [23,57,58]. MPs with $D/W < 0.5$, indicating LOF, appear most frequently in the meander strategy. This higher LOF tendency can be attributed to the sharp thermal gradients (Table 4) and continuous, unidirectional laser path, which create abrupt transitions between heated and unheated zones. Such conditions, compounded by possible instabilities (uneven powder distribution or insufficient overlap), result in localized under-penetration and LOF defects typical of conduction-dominated regimes.

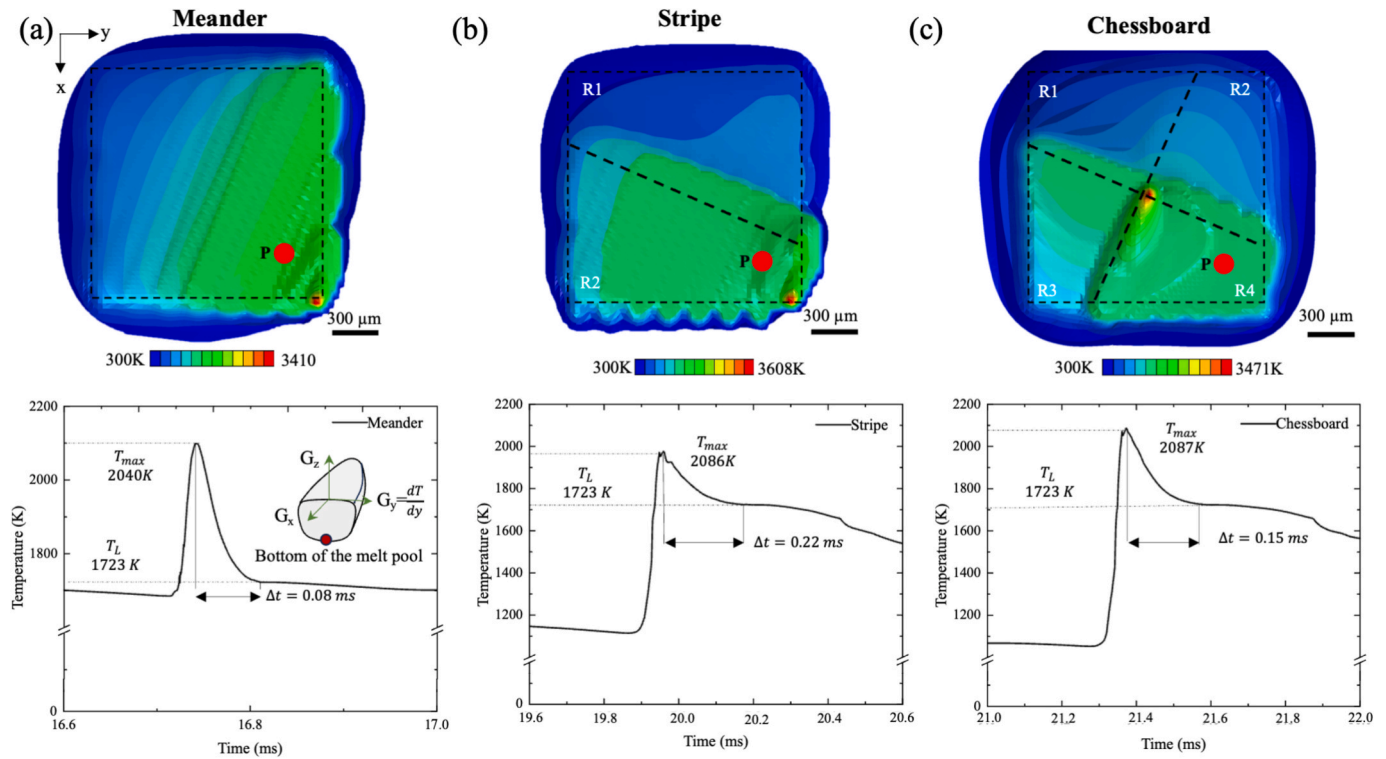


Fig. 5. Temperature–time curves at point P (bottom of MP) across a full layer: (a) meander, (b) stripe, (c) chessboard.

Table 4

Numerical cooling rates and temperature gradients at the top and bottom of the MP at point P.

	Meander		Stripe		Chessboard	
	Top	Bottom	Top	Bottom	Top	Bottom
Cooling Rate ($G \times R$) ($\times 10^5$ K/s)	3.2	4.85	1.1	1.93	1.6	2.8
Temperature Gradient (G) ($\times 10^6$ K/m)	10.06	28.85	12.5	21.35	13.5	22.6

By contrast, stripe and chessboard scans—owing to greater overlap and more uniform thermal fields—show reduced LOF incidence.

According to Fig. 6d, both the stripe and chessboard scanning strategies exhibit instances of transition and keyhole melting modes, in contrast to the meander strategy, indicating localized thermal intensification. MPs with depth-to-width (D/W) ratios between 1.0 and 1.5 fall within the transition regime, while those with D/W ratios exceeding 1.5 indicate the keyhole mode—characterized by deep, narrow MPs resulting from excessive energy input and multiple laser reflections within the MP. The lower temperature gradients calculated for the stripe and chessboard strategies (Table 4), attributed to their discontinuous scanning paths and higher overlap between adjacent scan regions (Fig. 4), may lead to localized heat accumulation. This accumulation can give rise to deeper MPs—particularly those in the keyhole mode—compared to the shallower conductive-mode MPs. In other words, if localized heating becomes excessive, it can surpass the material’s vaporization threshold, triggering keyhole mode melting. This melting regime is typically associated with increased porosity and structural defects. Therefore, while reduced temperature gradients in the stripe and chessboard strategies may enhance bonding uniformity, they also pose a trade-off by increasing the risk of keyhole-induced defects due to localized overheating.

Analysis of the MP types formed in the three samples (Fig. 6d) revealed that, despite the presence of multiple melting regimes, the conduction mode remains the most statistically dominant across all

scanning strategies. This mode, typically associated with moderate energy density, promotes stable MP formation and effective bonding between adjacent tracks and layers. The observed dominance of the conduction mode is consistent with the results of thermal simulations based on conduction-driven assumptions. However, a detailed understanding of the thermal influence of other MP modes, such as keyhole and transition on the primary conduction mode is necessary. The presence of these modes may introduce localized thermal fluctuations, potentially causing deviations in the microstructural features that evolve under different scanning strategies.

4.2. Implications of MP morphology for microstructural evolution

Differences in grain characteristics- including size, morphology, and crystallographic orientation- can be strongly influenced by the interactions between conductive MPs (the dominant MP type in all samples) and other MPs. In other words, interactions between two conductive MPs differ from those involving conductive-keyhole or conductive-transition MPs, potentially leading to deviations from the grain features typically expected under purely conductive conditions. To investigate these effects, EBSD analysis was performed ~ 0.7 mm below the last solidified layer in the LPBF-316LSS cubes (Fig. 7a). Notably, while conductive MPs remain the dominant mode across all strategies, at this depth, the accumulation of keyhole and transition MPs is more pronounced in the stripe and chessboard samples, respectively, while the meander sample is almost entirely composed of conductive MPs.

The EBSD IQ maps in Fig. 7a show the presence of high-angle grain boundaries (HAGBs, blue lines), low-angle grain boundaries (LAGBs, red lines), and MPs outlined by dashed black lines. Different MP modes—conductive (C), keyhole (K), and transition (T)—can be identified from their depth/width variations. The associated grain size distributions (Fig. 7b) reveal coarse grains ranging from 10–100 μm across all strategies, with average sizes of 38.2 μm (meander), 48.1 μm (stripe), and 61.8 μm (chessboard). The fractions of HAGBs and LAGBs are shown in Fig. 7c: the meander sample exhibits the highest HAGB fraction, consistent with its smaller average grain size, whereas the stripe

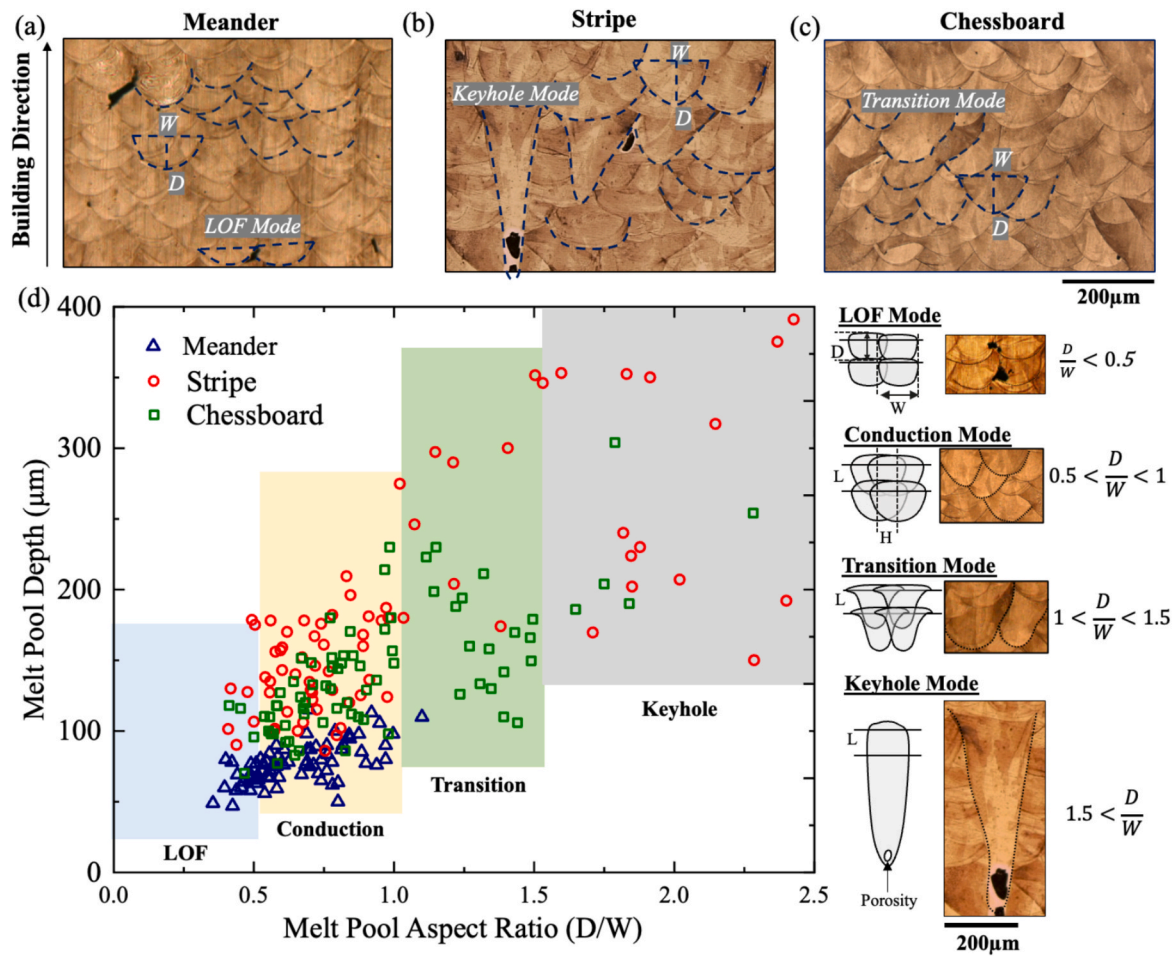


Fig. 6. Cross-section MPs for (a) meander, (b) stripe, (c) chessboard; (d) melting-mode distribution by D/W .

and chessboard samples show greater LAGB accumulation, with the chessboard sample exhibiting the highest fraction overall.

The prevalence of LAGBs in the stripe and chessboard samples reflects the development of fine substructural features within coarse grains. These features are associated with solidification sub-grains, dislocation-rich walls, and solute segregation driven by Marangoni convection and particle accumulation mechanisms[59]. Quantitative comparison of the highlighted areas (black rectangles in Fig. 7a) confirms that stripe and chessboard samples, which contain more frequent keyhole and transition MP interactions, accumulate more LAGBs than the meander sample, where MP interactions are almost exclusively conductive.

The inverse pole figure (IPF) maps along the building direction (BD) (Fig. 8a) highlight orientation differences across the three scan strategies. The corresponding [001], [110], and [111] pole figures (PFs) (Fig. 8b) show that, although the overall orientation distribution is broadly random, the accumulation of poles near the BD axis varies by strategy. In the meander sample, [110] poles are preferentially aligned with BD. In the stripe sample, orientations are closer to [100], with some tendency toward [111], whereas the chessboard sample exhibits stronger [111] alignment. The rotation of the preferred poles of grains parallel to the three axes of the samples is also depicted in the crystallographic orientation–axes plot in Fig. 8c. Accordingly, rotations of approximately 90° and 35.9° are estimated for the preferred orientations of grains parallel to the BD axis when transitioning from the meander to the stripe and chessboard scan strategies, respectively.

The deviations can be attributed to the interactions between different MPs mode, marked with black arrows in Fig. 8a. The MP modes govern the migration behavior of the solid–liquid interface. In meander,

conduction-dominated MPs reduce curvature at the pool bottom, resulting in vertical solid–liquid interface migration and epitaxial grain growth along BD. This explains why elongated grains are evident in Fig. 8a and why orientations align with the maximum thermal gradient, even if this corresponds to [110] rather than the typical FCC [100] growth direction. This deviation may arise because, during crystal growth, grains that align with the maximum temperature gradient are retained throughout the solidification process, even if their crystallographic orientations do not align with the easiest growth direction [60].

The presence of keyhole and transition MPs in the stripe and chessboard strategies also influences the growth direction of coarse grains in these samples. The more pronounced curvature at the bottom of the MPs in keyhole and transition modes, compared to the conductive mode, shifts the solid–liquid interface laterally at the bottom of these MPs [25]. This results in the lateral growth of grains relative to the BD axis. The lateral shift of the solid–liquid interface at the bottom of these MPs re-directs the thermal gradient away from BD, promoting lateral epitaxial growth and orientation deviations (marked with black arrows in Fig. 8a).

The MP shape, which largely determines the orientation of the thermal gradient (normal to the MP boundary), plays a key role in the development of crystallographic orientations in three samples. In the meander sample, interactions between conductive MPs reduced the curvature of MPs, which promoted frequent migration of the solid–liquid interface toward the BD. This migration aligned with the semi-circular curvature of conductive MPs, resulting in grain growth along the thermal gradient to some extent parallel to BD. By contrast, when transition or keyhole MPs interacted with conductive MPs, the solid–liquid interface became more unstable and exhibited greater curvature.

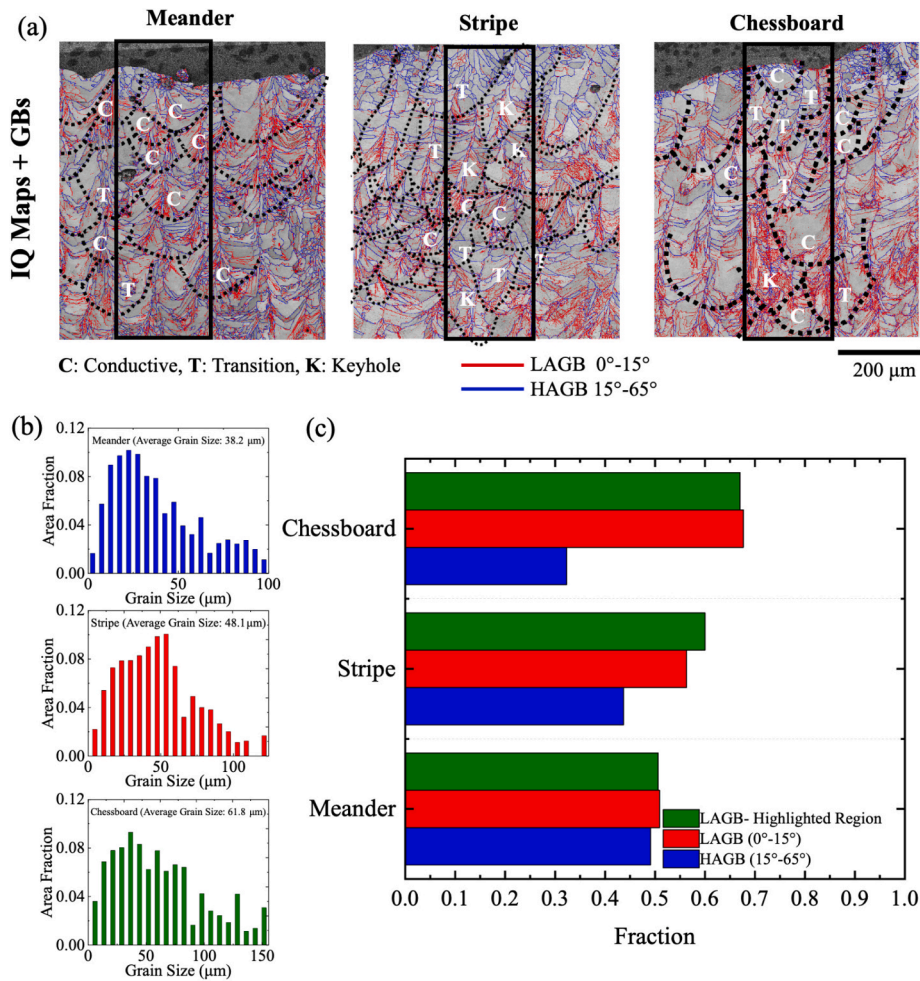


Fig. 7. (a) EBSD IQ maps with HAGBs (blue) and LAGBs (red), showing MPs (dashed lines) for meander, stripe, and chessboard; (b) grain size distributions; (c) HAGB/LAGB fractions in the selected regions. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

This led to lateral migration of the interface, accompanied by fluctuations in the thermal gradient and shifts in the location of maximum thermal gradients. Such fluctuations redirected the preferred crystallographic growth orientations [25,61,62]. In other words, different types of MP interactions modify the melt pool curvature, thereby altering the direction of the local maximum thermal gradient, which in turn dictates the preferential crystallographic orientations during solidification.

To evaluate the combined effects of different MP modes on crystallographic orientations, grain morphology, and size, EBSD crystal direction maps were generated for the three samples (Fig. 9a). Specific overlapping regions were highlighted with white circles: (i) conductive–conductive MP overlap (C2 and C3) in the meander sample, (ii) a transition–conductive MP interaction (T and C1) in the stripe sample, and (iii) a keyhole–conductive MP interaction (K and C2) in the chessboard sample. These regions are magnified in Fig. 9b. Crystallographic orientations within the conductive MPs were further quantified along four directions (left/right transverse edges, top, and bottom), indicated by dashed red arrows in Fig. 9a, and the distributions are presented in Fig. 9c.

From Figs. 9a,b it can be inferred that in the meander sample, remelting of C3 by the adjacent C2 shifts the grain growth direction at the top of C3—from [310]/[221] toward [311]—producing a morphological transition from equiaxed to columnar grains. In the stripe sample, the transition MP (T) above C1 enables coarse grains originating from C1 to extend through the entire transition MP. In the chessboard sample, overlap between K and C2 transforms the relatively columnar

grains on the right side of C2 into finer, more equiaxed grains within the keyhole MP.

The orientation accumulations in Fig. 9c shows that the dominant [310] direction in the conductive MPs of the meander sample rotates to [211] and [311] in the stripe and [211] in the chessboard samples, suggesting that interactions with the other MP types influence the crystallographic directions of the conductive MPs. While overall orientation distributions remain broadly random, these rotations indicate that keyhole and transition MPs influence the crystallographic orientations of nearby conductive MPs. In meander, where conductive MPs dominate, orientations remain closer to [310] (slightly deviated from [001]), whereas stripe and chessboard show stronger deviations toward [111].

These results can be rationalized by thermodynamic solidification conditions and orientation selection mechanisms. Conductive–conductive MP interactions produce shallow, semi-elliptical geometries, giving lower G/R values and less elongated grains. By contrast, interactions between conductive MPs and transition/keyhole MPs increase melt pool curvature and raise G/R, promoting more elongated grains in stripe and chessboard. Transition MPs, with their curved boundaries, locally tilt the gradient laterally, extending coarse grains into adjacent regions and deviating orientations from the expected [100] FCC growth direction. Keyhole MPs, with deep and narrow geometries, further enhance G/R and alter grain morphology. Thus, changes in melt pool curvature and interface migration caused by different MP interactions directly modify the direction of the maximum

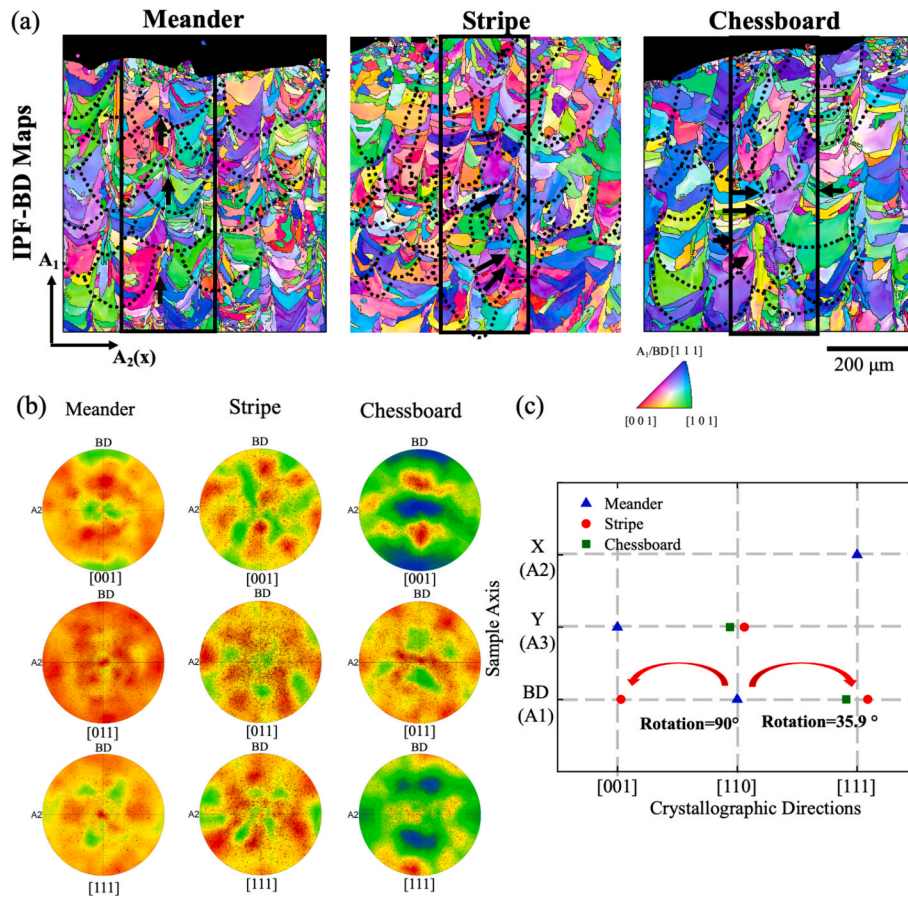


Fig. 8. (a) IPF-BD maps showing crystallographic orientations; (b) pole figures; (c) orientation-axes plots for meander, stripe, and chessboard strategies.

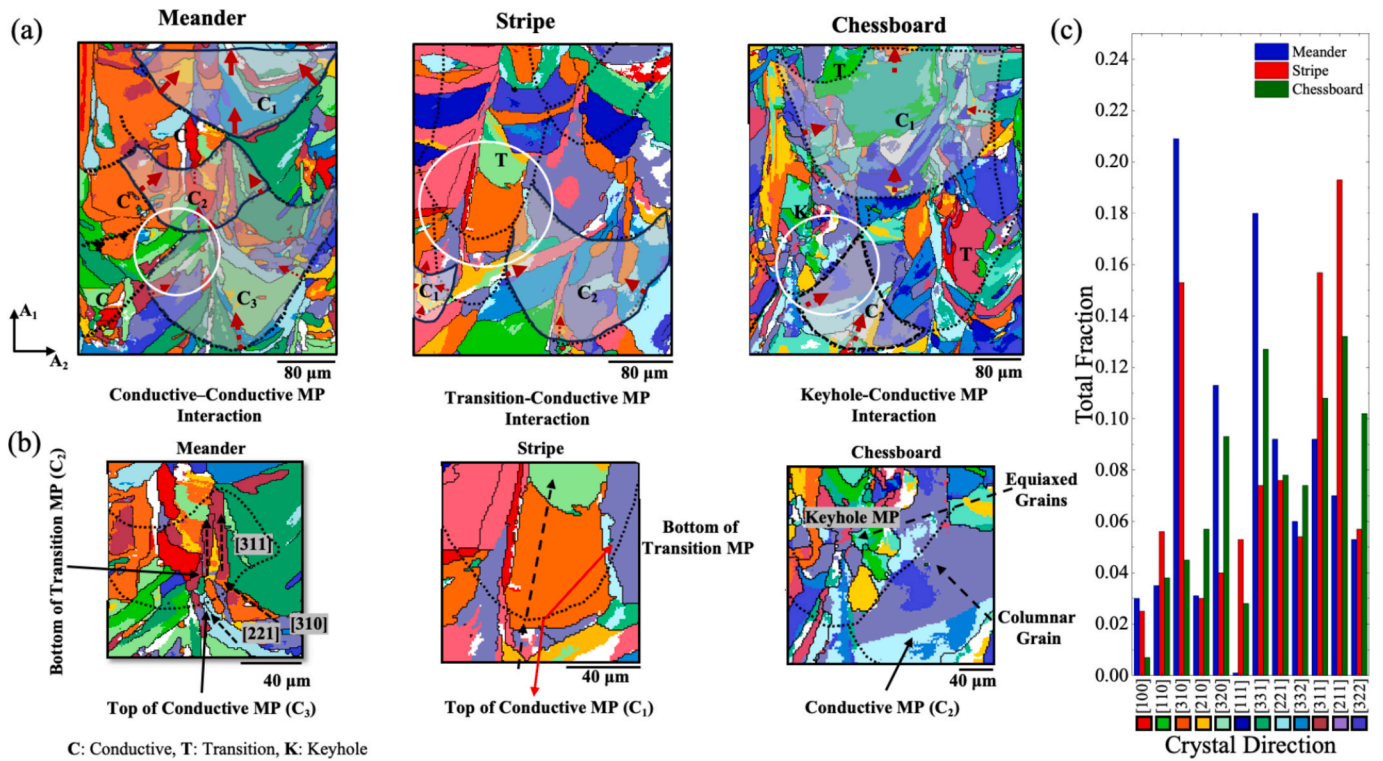


Fig. 9. (a) EBSD crystal direction maps highlighting MP interactions; (b) magnified maps of selected overlapping regions (white circles); (c) orientation distributions along transverse and vertical directions of conductive MPs.

thermal gradient, explaining the observed orientation rotations toward [111] and [211] in stripe and chessboard.

4.3. In-Situ heat treatment and thermal histories influenced by MP interactions

The size, morphology, and orientation of coarse grains are also significantly influenced by reheating and re-cooling during the LPBF process in the meander, stripe, and chessboard samples. These effects are induced by adjacent MPs formed during laser scan tracking and the deposition of subsequent layers (e.g., L_{N+1}) over previous layers (e.g., L_N) [63,64]. These interactions can lead to remelting in the upper and adjacent regions of the MPs, resulting in the formation of heat-affected zones (HAZs) around the MPs [64]. It is noted that the extent of remelting and the development of HAZs depend on the scanning pattern and the degree of interaction between different types of MPs, which can influence grain size and morphology by activating restoration mechanisms and inducing deviations in preferred crystallographic orientations. To gain deeper insights into the in-situ heat treatments induced by remelting and HAZ formation during the LPBF process, a multi-layer simulation was performed over the five top layers.

To calculate the peak temperatures during laser movement—corresponding to the scanning patterns—the temperature history at point P located at the top of MP was tracked across the five layers below the top surface as they were sequentially deposited along the BD (Fig. 10a). As the second layer was fabricated, point P in the first layer experiences a similar cycle of preheating, melting, and post-heating, but with lower peak temperatures. The intentional remelting of point P during the construction of the second layer aims to create a metallurgically sound interface between consecutive build layers in the LPBF process.

Although the interlayer delay time of 12 s (as specified by the GE machine in this study) minimizes heat accumulation, the thermal patterns in subsequent layers differed across the three scan strategies. In the meander sample (Fig. 10b), during the fabrication of the second layer, the temperature at point P increased to 1738 K, which was above the liquidus temperature of 1723 K, and it reached 1478 K during the deposition of the third layer. The subsequent melting pattern diagram for the meander strategy confirms that although the second layer reached a temperature above the liquidus temperature, it was not sufficient to significantly expand the depth of the MP. The multi-layer thermal analysis of the stripe strategy (Fig. 10c) reveals that point P undergoes repeated melting after the deposition of the second and third layers. This recurring thermal influence results in a larger MP or keyhole formation in some areas. A comparable pattern was observed in the chessboard samples (Fig. 10d); however, by the third cycle, the temperature at point P approaches the liquidus temperature, indicating a nearly complete melting process.

Accordingly, the peak temperature–depth profiles for the three scan strategies are presented in Fig. 11a, together with the pseudo-binary phase diagram of LPBF-316LSS (Fig. 11b) and the corresponding phase maps from region ~ 0.7 mm below the top surface (Fig. 11c). These results illustrate how the thermal field generated at the surface penetrates into the underlying layers, effectively representing the temperature gradient beneath the melt pool. At the surface, maximum temperatures reach ~ 3608 K for the stripe strategy and ~ 3471 K for the chessboard strategy, both well above the liquidus temperature (1723 K). This thermal field extends up to ~ 150 μm into the material, indicating partial remelting of previously solidified layers in stripe and chessboard, whereas in the meander strategy remelting is confined to ~ 50 μm .

Beyond ~ 150 μm , peak temperatures drop below 1473 K, consistent with conditions for recovery and recrystallization, and below ~ 973 K

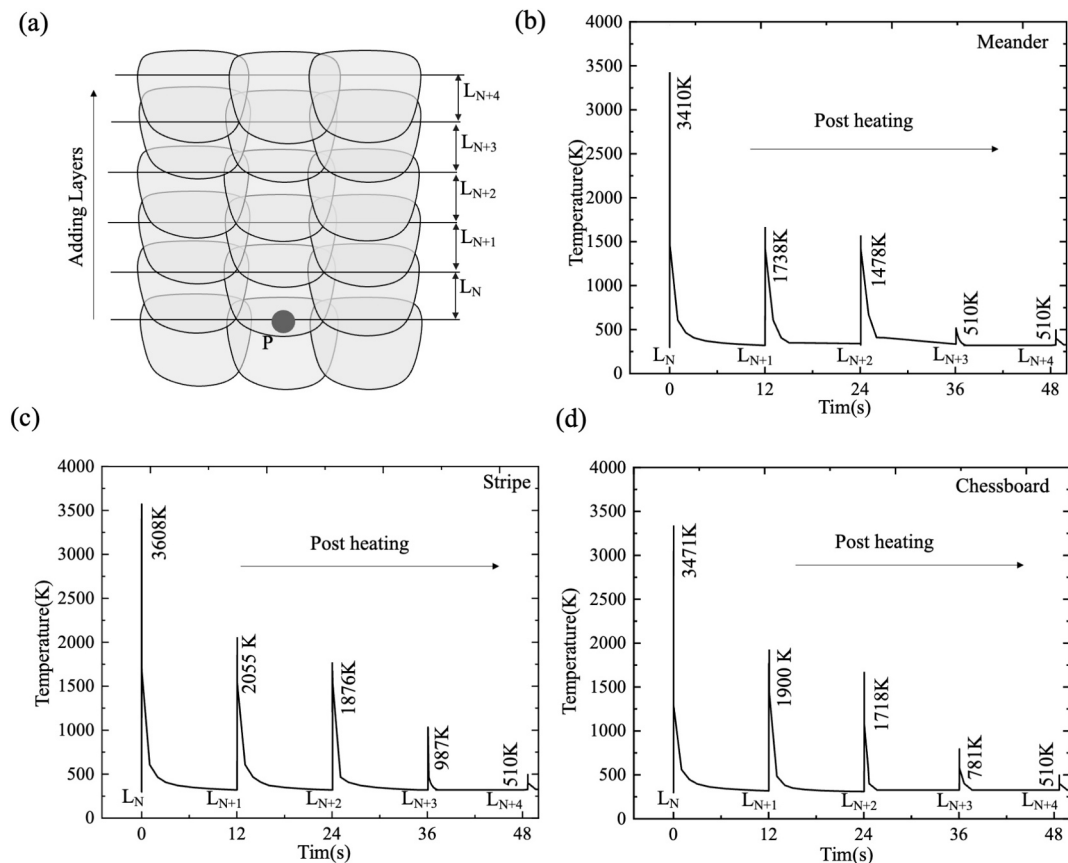


Fig. 10. (a) Schematic of thermal history tracking at point P; (b–d) simulated temperature profiles across successive layers for meander, stripe, and chessboard strategies.

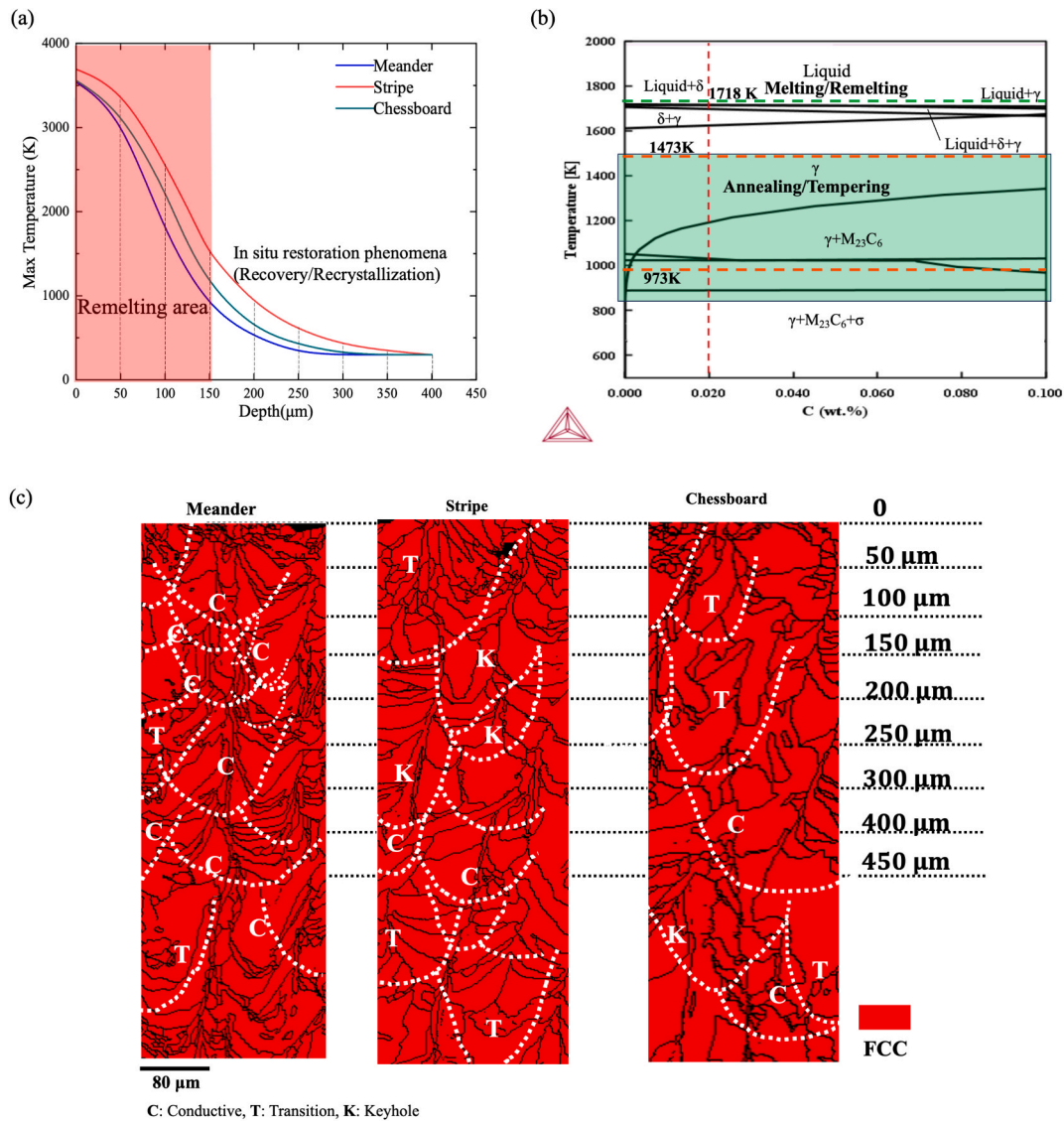


Fig. 11. (a) Peak temperature versus depth for meander, stripe, and chessboard; (b) pseudo-binary phase diagram of LPBF-316LSS; (c) EBSD phase maps of regions highlighted in Figs. 7a and 8a.

the FCC austenite phase remains stable. The phase diagram (Fig. 11b) and phase maps (Fig. 11c) together confirm that austenite is the dominant stable phase across these regions. Layers exposed to intermediate temperatures between 1473 K and 973 K may undergo homogenization of alloying elements and activation of restoration mechanisms such as recrystallization and recovery. These processes reduce dislocation density and promote the formation of finer, more equiaxed grains.

Therefore, the in-situ heat treatments induced during the LPBF process—combined with varying degrees of remelting and overlap among conduction, transition, and keyhole MPs—are key factors governing the observed shifts in grain size, morphology, and crystallographic orientation.

The occurrence of in-situ recrystallization and recovery is also evident in the KAM maps and corresponding KAM values shown in Fig. 12a–b. In these regions, the accumulation of KAM values between 1° and 2°, as well as the variation in average KAM values with increasing distance from the top surface, can be observed. Regions located ~ 100, 200, 300, and 400 μm below the top surface exhibit lower KAM values compared to the overall average, as illustrated in the KAM–depth plots (Fig. 12c). This trend indicates the activation of dislocation annihilation mechanisms in deeper layers. However, repeated reheating and re-

cooling during LPBF can also promote remelting in overlapping regions, leading to the formation of new grains and, in some cases, an increase in dislocation density. Thus, while certain regions show lower KAM values due to recovery, others display higher values where remelting and grain overlap occur.

The accumulation of KAM values reflects dislocation density, either from misfit dislocations or geometrically necessary dislocations (GNDs) generated by thermal gradients [65]. The observed reduction of KAM respect to average value with depth highlights the role of recovery in mitigating dislocation buildup. In contrast, the chessboard sample shows persistently high KAM values at greater depths, differing from the meander and stripe samples. These elevated values are particularly concentrated in regions where conduction, transition, and keyhole MPs intersect, as marked by the black circle in Fig. 12a. This observation is consistent with the higher LAGB density in the chessboard sample (Fig. 7a), confirming that MP interactions enhance local dislocation accumulation.

The conduction, transition, and keyhole melt pool (MP) modes differ in penetration depth and lateral overlap, which strongly influences the extent of partial remelting and the formation of heat-affected zones (HAZs). In the meander sample, conduction MPs dominate in layers

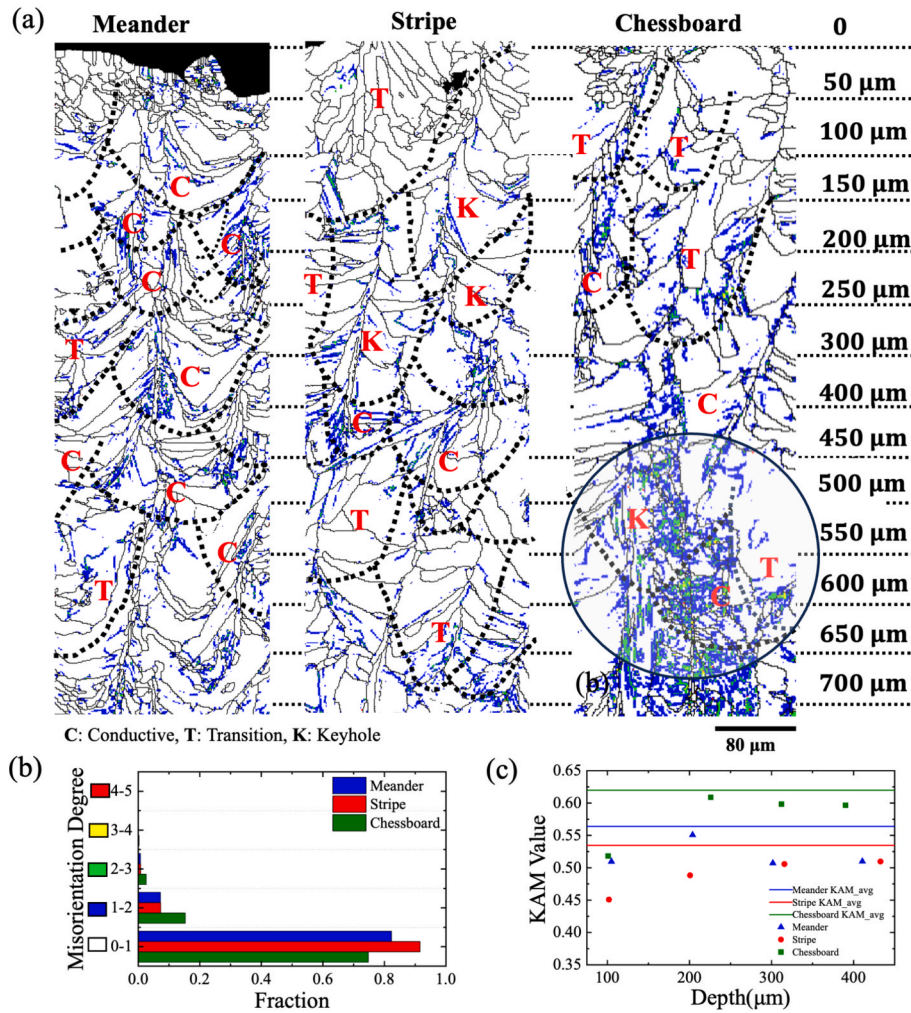


Fig. 12. (a) KAM maps of selected regions; (b) KAM value, (c) corresponding KAM–depth plots showing variations in dislocation accumulation with depth.

located ~ 0.7 mm below the top surface (Fig. 7a). Because their penetration depth is shallower and wider than that of transition or keyhole MPs, the resulting remelting of adjacent and underlying MPs is relatively limited, producing narrower HAZ regions. By contrast, in the stripe and chessboard samples, the prevalence of transition and keyhole MPs generates deeper and narrower remelting zones.

Such geometric differences in partial remelting directly affect local thermal conditions. Overlap among different MP types enhances dislocation accumulation by promoting the formation of geometrically necessary dislocations (GNDs), while annihilation processes may still act in inner layers to reduce dislocation density. Variations in melt pool size, morphology, and crystallographic orientation therefore alter the local temperature gradient and cooling rate, ultimately modifying grain size, morphology, and preferred crystallographic orientations [66,67].

4.4. MP interactions and their effects on grain characteristics

Grain shape aspect ratio (GSAR) maps and the corresponding values for the three different scanning strategies (Fig. 13) show that the meander sample has the lowest minor/major axis ratio (<0.33), indicating a higher fraction of finer, less elongated grains. This is consistent with the greater accumulation of conduction MPs in meander compared to stripe and chessboard, where deeper and narrower MPs in keyhole and transition modes are more prevalence. The presence of these modes promotes the formation of larger columnar grains, as highlighted in the chessboard sample (black circle in Fig. 13).

This effect is observed both when keyhole or transition MPs partially

remelt conduction MPs and when they overlap with each other. By contrast, in the meander sample, frequent interactions between conduction MPs combined with in-situ recrystallization and recovery processes favor the development of smaller, less elongated grains. Overall, grain evolution in LPBF is governed by two key factors: (i) solidification conditions—particularly the rapid cooling rates and applied thermal gradients that depend on MP morphology—and (ii) partial remelting and intrinsic heat treatments during reheating and cooling of successive layers, further influenced by laser transverse motion and hatch spacing [66].

4.5. Orientation solidification map induced by scan strategy–MP dynamics

Variations in scanning patterns influence the type and size of MPs, the extent of in-situ heat treatment, and interactions between conductive, keyhole, and transition MPs. These factors directly affect grain size, morphology, and crystallographic orientations. Their combined effects are summarized in the orientation solidification map shown in Fig. 14a.

The G/R and $G \times R$ values were estimated from FE simulation at four representative locations within conductive MPs—the two transverse edges, the top, and the bottom (Fig. 14b). Crystallographic orientations of grains within these MPs were then compared using IPF maps along the BD axis.

According to the solidification map (Fig. 14a), the meander strategy exhibits the highest cooling rates ($G \times R$ up to 4.9×10^6 K/s) and relatively low G/R values ($32\text{--}60 \times 10^6$ K·s/m²). Spatial variations

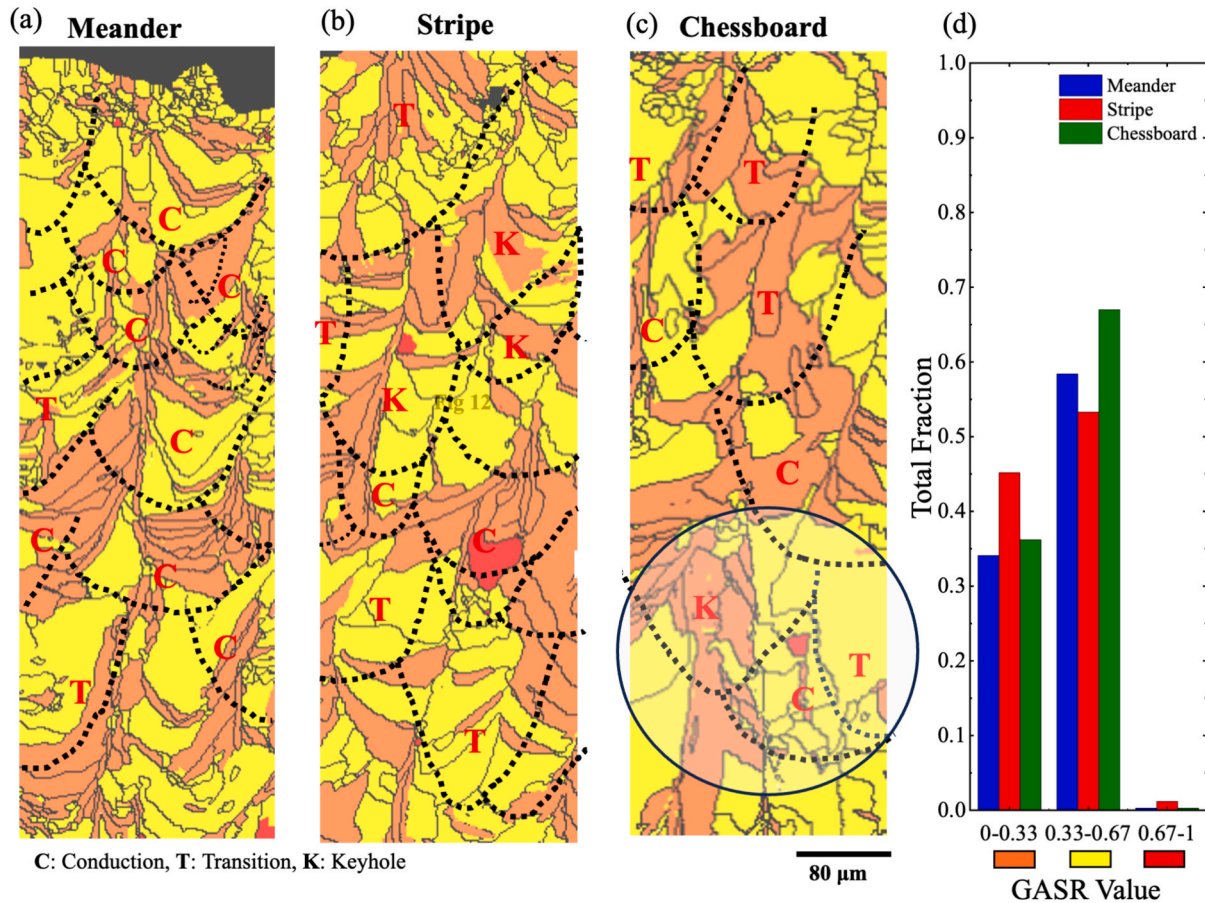


Fig. 13. (a) GSAR maps for selected regions; (b) corresponding GSAR values for meander, stripe, and chessboard samples.

between the four MP positions are also smaller in meander than in stripe or chessboard, producing more uniform solidification conditions. These factors, together with higher $G \times R$, promote finer and less elongated grains. In contrast, stripe and chessboard show higher G/R but lower $G \times R$ values, favoring coarser and more elongated grains. Notably, the top-bottom differences in G/R and $G \times R$ are largest for stripe, explaining its stronger tendency toward elongated grains. These predictions are consistent with the GSAR values (Fig. 13) and grain size distributions (Fig. 7b).

In meander MPs, the preferred crystallographic orientation shifts toward directions between [111] and [101], with minimal deviation from [101], as seen in the IPF-BD indices (Fig. 14a). This is consistent with PF results (Fig. 8b) and orientation accumulations in Fig. 9c. The repeated formation of shallow conduction MPs stabilizes orientations near BD even when they shift the orientation from the [310] direction (which is close to [001]) toward a direction between [111] and [101] directions.

In stripe, conductive MPs exhibit preferred orientations closer to [111] and [100], as shown in Fig. 14a, matching the PFs in Fig. 8b. The influence of keyhole MPs leads to additional rotations, particularly toward [211], as seen in Fig. 9. These interactions redirect crystallographic growth orientations away from purely conductive-driven directions.

Similarly, in chessboard, conduction MPs show orientations concentrated near [111] (Fig. 14a and Fig. 8b). Transition MPs further shift orientations toward [211] (Fig. 9), but the combined effect ultimately stabilizes grain orientations around [111]. Compared to meander, where conduction MPs dominate, the higher frequency of transition MPs in chessboard increases deviations from [100] but promotes elongated grains aligned near [111].

Fig. 14c schematically summarizes these mechanisms. In meander, conduction MPs overlap shallowly, leading to stable orientations near [100] and promoting finer equiaxed grains through enhanced restoration processes. In stripe, keyhole MPs dominate, driving deviations toward [210]/[211] and [111], accompanied by coarser elongated grains. In chessboard, transition MPs shift orientations toward [111] and [211], altering remelting depth and HAZ extent. Thus, the orientation map demonstrates how MP dynamics, governed by scan strategy, control crystallographic orientation, grain morphology, and microstructural refinement.

5. Conclusions

This study investigated the influence of laser scanning strategies on thermal behavior of MP geometry, and microstructural evolution in LPBF-316LSS. By integrating a validated finite element framework with experimental characterization, a novel orientation solidification map was developed that correlates local solidification conditions with crystallographic orientations, providing a predictive tool for microstructure control in LPBF. The main findings are:

The meander scan strategy exhibited the highest cooling rates ($3-5 \times 10^6$ K/s) and temperature gradients ($10-30 \times 10^6$ K/m), leading to the formation of smaller, shallower MPs. In contrast, the stripe pattern produced the lowest values, resulting in wider and deeper MPs. Multilayer thermal simulations revealed partial remelting through three layers (~ 150 μm) in the stripe strategy and two layers (~ 100 μm) in the chessboard strategy, while the meander pattern exhibited minimal remelting (~ 50 μm), indicating more stable thermal profiles.

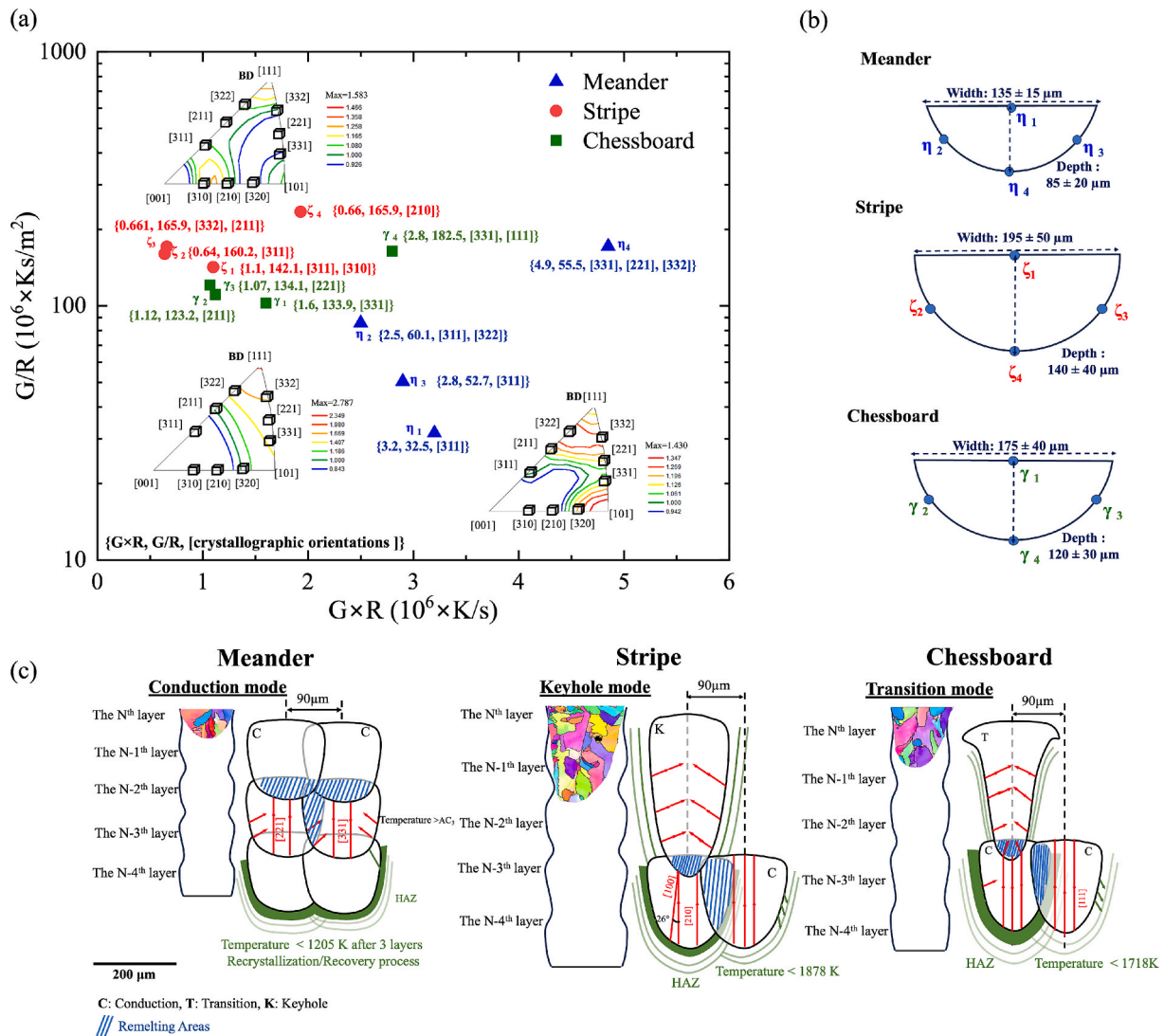


Fig. 14. (a) Orientation–solidification map showing cooling rate ($G \times R$), solidification conditions (G/R), and IPF-BD indices at four locations within conductive MPs; (b) schematic of MP geometry and measurement points; (c) schematic of grain growth directions under conduction, transition, and keyhole modes for different scanning strategies.

Conduction mode dominated across all strategies, but differences in depth-to-width (D/W) ratios distinguished their susceptibility to defects. Meander showed occasional lack-of-fusion due to low D/W, stripe was prone to keyhole formation, and chessboard displayed a higher frequency of transition MPs—highlighting the role of scanning strategy in local energy input.

LPBF microstructure was shown to be hierarchical, with scan strategy influencing not only grain morphology and texture but also grain boundary character. High cooling rates in meander promoted HAGBs and refinement, while the lower cooling rates and extended remelting in stripe and chessboard encouraged sub-grain formation and higher LAGB fractions.

Differences in MP penetration depths dictated the geometry of partial remelting and HAZ formation. Conductive MPs produced shallower and wider remelting zones, while keyhole and transition MPs generated deeper, narrower zones. These variations altered local cooling rates and gradients, leading to distinct grain morphologies. Meander showed the lowest KAM ($\sim 0.42^\circ$) and more equiaxed grains, while chessboard displayed the highest KAM ($\sim 0.73^\circ$) and columnar grains, with stripe falling in between.

The orientations of conductive MPs shifted depending on their interactions with other MP types. In meander, dominated by

conduction MPs, the prevalent [310] orientation remained close to [001]. In stripe and chessboard, frequent keyhole and transition MPs shifted orientations toward [211] and [311], indicating stronger deviations toward [111]. Although the overall texture remained largely random, these interactions clearly influenced orientation selection.

The orientation solidification map linked thermal gradients (G), solidification rates (R), and crystallographic orientations, providing a framework to explain strategy-dependent deviations. Meander consistently showed the highest $G \times R$ values, favoring finer grains and dominant [110]/[101] orientations. Stripe and chessboard, with lower $G \times R$, promoted coarser grains and orientations shifting toward [100]/[111] and [111], respectively. This map demonstrates how MP dynamics, governed by scan strategy, drive grain morphology and texture evolution, and serves as a predictive tool for microstructure-aware process design.

CRediT authorship contribution statement

Foroozan Forooghi: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ayda Shahriari:** Writing – review & editing, Writing

– original draft, Investigation, Formal analysis, Data curation, Conceptualization. **Parisa Moazzen**: Writing – review & editing, Formal analysis, Data curation. **Mohsen Keshavarzan**: Writing – review & editing, Formal analysis, Data curation. **Nana Ofori-Opoku**: Writing – review & editing, Supervision, Conceptualization. **Mohsen Mohammadi**: Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- [1] D.C. Hofmann, S. Roberts, R. Otis, J. Kolodziejska, R.P. Dillon, J. Suh, A.A. Shapiro, Z.-K. Liu, J.-P. Borgonia, Developing gradient metal alloys through radial deposition additive manufacturing, *Sci. Rep.* 4 (2014) 5357, <https://doi.org/10.1038/srep05357>.
- [2] C. Körner, Additive manufacturing of metallic components by selective electron beam melting - a review, *Int. Mater. Rev.* 61 (2016) 361–377, <https://doi.org/10.1080/09506608.2016.1176289>.
- [3] D. Dev Singh, T. Mahender, A. Raji Reddy, Powder bed fusion process: a brief review, *Mater. Today Proc.* 46 (2021) 350–355, <https://doi.org/10.1016/j.matpr.2020.08.415>.
- [4] U. Scipioni Bertoli, G. Guss, S. Wu, M.J. Matthews, J.M. Schoenung, In-situ characterization of laser-powder interaction and cooling rates through high-speed imaging of powder bed fusion additive manufacturing, *Mater Des* 135 (2017) 385–396, <https://doi.org/10.1016/j.matdes.2017.09.044>.
- [5] T. Keller, G. Lindwall, S. Ghosh, L. Ma, B.M. Lane, F. Zhang, U.R. Kattner, E.A. Lass, J.C. Heigel, Y. Idell, M.E. Williams, A.J. Allen, J.E. Guyer, L.E. Levine, Application of finite element, phase-field, and CALPHAD-based methods to additive manufacturing of Ni-based superalloys, *Acta Mater.* 139 (2017) 244–253, <https://doi.org/10.1016/j.actamat.2017.05.003>.
- [6] M.M. Kirka, K.A. Unocic, N. Raghavan, F. Medina, R.R. Dehoff, S.S. Babu, Microstructure development in electron beam-melted Inconel 718 and associated tensile properties, *JOM* 68 (2016) 1012–1020, <https://doi.org/10.1007/s11837-016-1812-6>.
- [7] M.V. Pantawane, Y.-H. Ho, S.S. Joshi, N.B. Dahotre, Computational assessment of thermokinetics and associated microstructural evolution in laser powder bed fusion manufacturing of Ti6Al4V Alloy, *Sci. Rep.* 10 (2020) 7579, <https://doi.org/10.1038/s41598-020-63281-4>.
- [8] M. Mahmoudi, G. Tapia, K. Karayagiz, B. Franco, J. Ma, R. Arroyave, I. Karaman, A. Elwany, Multivariate calibration and experimental validation of a 3D finite element thermal model for laser powder bed fusion metal additive manufacturing, *Integr Mater Manuf Innov* 7 (2018) 116–135, <https://doi.org/10.1007/s40192-018-0113-z>.
- [9] M. Amoozezaei, S. Gurevich, N. Provatas, Orientation selection in solidification patterning, *Acta Mater.* 60 (2012) 657–663, <https://doi.org/10.1016/j.actamat.2011.10.006>.
- [10] P. Biswas, J. Ma, Development of crystallographic misorientation in laser powder bed fusion 316L stainless steel, *Addit. Manuf.* 80 (2024) 103951, <https://doi.org/10.1016/j.addma.2023.103951>.
- [11] J. Zou, Y. Gaber, G. Voulazeris, S. Li, L. Vazquez, L.F. Liu, M.Y. Yao, Y.J. Wang, M. Holynski, K. Bongs, M.M. Attallah, Controlling the grain orientation during laser powder bed fusion to tailor the magnetic characteristics in a Ni-Fe based soft magnet, *Acta Mater.* 158 (2018) 230–238, <https://doi.org/10.1016/j.actamat.2018.07.064>.
- [12] O. Gokcekaya, T. Ishimoto, S. Hibino, J. Yasutomi, T. Narushima, T. Nakano, Unique crystallographic texture formation in Inconel 718 by laser powder bed fusion and its effect on mechanical anisotropy, *Acta Mater.* 212 (2021) 116876, <https://doi.org/10.1016/j.actamat.2021.116876>.
- [13] A.T. Polonsky, W.C. Lenthe, M.P. Echlin, V. Livescu, G.T. Gray, T.M. Pollock, Solidification-driven orientation gradients in additively manufactured stainless steel, *Acta Mater.* 183 (2020) 249–260, <https://doi.org/10.1016/j.actamat.2019.10.047>.
- [14] H. Qin, V. Fallah, Q. Dong, M. Brochu, M.R. Daymond, M. Gallyneault, Solidification pattern, microstructure and texture development in Laser Powder Bed Fusion (LPBF) of Al10SiMg alloy, *Mater Charact* 145 (2018) 29–38, <https://doi.org/10.1016/j.matchar.2018.08.025>.
- [15] Y. He, B. Webber, A solidification mode selection process map for laser powder bed fusion additive manufacturing of B-modified Ti6Al4V, *Addit. Manuf.* 59 (2022) 103086, <https://doi.org/10.1016/j.addma.2022.103086>.
- [16] F.H. Kim, H. Yeung, E.J. Garboczi, Characterizing the effects of laser control in laser powder bed fusion on near-surface pore formation via combined analysis of in-situ melt pool monitoring and X-ray computed tomography, *Addit. Manuf.* 48 (2021) 102372, <https://doi.org/10.1016/j.addma.2021.102372>.
- [17] S. Chowdhury, N. Yadaiah, C. Prakash, S. Ramakrishna, S. Dixit, L.R. Gupta, D. Buddhi, Laser powder bed fusion: a state-of-the-art review of the technology, materials, properties & defects, and numerical modelling, *J. Mater. Res. Technol.* 20 (2022) 2109–2172, <https://doi.org/10.1016/j.jmrt.2022.07.121>.
- [18] K. Deshmukh, A. Riensche, B. Bevans, R.J. Lane, K. Snyder, H. (Scott) Halliday, C. B. Williams, R. Mirzaeifar, P. Rao, Effect of processing parameters and thermal history on microstructure evolution and functional properties in laser powder bed fusion of 316L, *Mater Des* 244 (2024) 113136, <https://doi.org/10.1016/j.matdes.2024.113136>.
- [19] K. Dai, X. He, W. Zhang, D. Kong, R. Guo, M. Hu, K. He, C. Dong, Tailoring the microstructure and mechanical properties for Hastelloy X alloy by laser powder bed fusion via scanning strategy, *Mater. Des.* 235 (2023) 112386, <https://doi.org/10.1016/j.matdes.2023.112386>.
- [20] L. Zhou, Z. Peng, J. Chen, Y. Ren, Y. Niu, W. Qiu, J. Tang, Z. Li, C. Li, The evolution of microstructure and mechanical properties of Inconel 625 alloy fabricated by laser powder bed fusion via novel hybrid scanning strategy, *Mater. Sci. Eng. A* 911 (2024) 146925, <https://doi.org/10.1016/j.msea.2024.146925>.
- [21] T. Kaleli Alay, M. Cagirci, M.Y. Yalcin, A. Yagmur, S. Tirkes, E. Aydogan, C.H. Gur, Tailoring the microstructure and mechanical properties of IN718 alloy via a novel scanning strategy implemented in laser powder bed fusion, *Materials Science and Engineering: A* 884 (2023) 145543, <https://doi.org/10.1016/j.msea.2023.145543>.
- [22] N. Nadammal, T. Mishurova, T. Fritsch, I. Serrano-Munoz, A. Kromm, C. Haberland, P.D. Portella, G. Bruno, Critical role of scan strategies on the development of microstructure, texture, and residual stresses during laser powder bed fusion additive manufacturing, *Addit. Manuf.* 38 (2021), <https://doi.org/10.1016/j.addma.2020.101792>.
- [23] H. Ali, H. Ghadbeigi, K. Mumtaz, Effect of scanning strategies on residual stress and mechanical properties of Selective Laser Melted Ti6Al4V, *Mater. Sci. Eng. A* 712 (2018) 175–187, <https://doi.org/10.1016/j.msea.2017.11.103>.
- [24] S.M.H. Hojjatzadeh, N.D. Parab, W. Yan, Q. Guo, L. Xiong, C. Zhao, M. Qu, L. I. Escano, X. Xiao, K. Fezzaa, W. Everhart, T. Sun, L. Chen, Pore elimination mechanisms during 3D printing of metals, *Nat. Commun.* 10 (2019) 3088, <https://doi.org/10.1038/s41467-019-10973-9>.
- [25] S.A. Khairallah, A.T. Anderson, A. Rubenchik, W.E. King, Laser powder-bed fusion additive manufacturing: physics of complex melt flow and formation mechanisms of pores, spatter, and denudation zones, *Acta Mater.* 108 (2016) 36–45, <https://doi.org/10.1016/j.actamat.2016.02.014>.
- [26] G.P. Dinda, A.K. Dasgupta, J. Mazumder, Texture control during laser deposition of nickel-based superalloy, *Scr. Mater.* 67 (2012) 503–506, <https://doi.org/10.1016/j.scriptamat.2012.06.014>.
- [27] K. Zeng, D. Pal, B. Stucker, A review of thermal analysis methods in Laser Sintering and Selective Laser Melting, n.d.
- [28] W. Di, Y. Yongqiang, S. Xubin, C. Yonghua, Study on energy input and its influences on single-track, multi-track, and multi-layer in SLM, *Int. J. Adv. Manuf. Technol.* 58 (2012) 1189–1199, <https://doi.org/10.1007/s00170-011-3443-y>.
- [29] S. Kolosov, E. Boillat, R. Glardon, P. Fischer, M. Locher, 3D FE simulation for temperature evolution in the selective laser sintering process, *Int. J. Mach. Tool Manuf.* 44 (2004) 117–123, <https://doi.org/10.1016/j.ijmachtools.2003.10.019>.
- [30] W. Yan, W. Ge, Y. Qian, S. Lin, B. Zhou, W.K. Liu, F. Lin, G.J. Wagner, Multi-physics modeling of single/multiple-track defect mechanisms in electron beam selective melting, *Acta Mater.* 134 (2017) 324–333, <https://doi.org/10.1016/j.actamat.2017.05.061>.
- [31] H. Chen, W. Yan, Spattering and denudation in laser powder bed fusion process: multiphase flow modelling, *Acta Mater.* 196 (2020) 154–167, <https://doi.org/10.1016/j.actamat.2020.06.033>.
- [32] P. Seraj, A. Akbarzadeh, M. Mohammadi, Finite element heat transfer analysis in selective laser melting of AlSi10Mg alloy using bi-directional scan strategy: enhanced thermal conductivity considerations, *J. Mater. Res. Technol.* 27 (2023), <https://doi.org/10.1016/j.jmrt.2023.11.067>.
- [33] S. Kollmannsberger, M. Carraturo, A. Reali, F. Auricchio, Accurate prediction of melt pool shapes in laser powder bed fusion by the non-linear temperature equation including phase changes, *Integr Mater Manuf Innov* 8 (2019) 167–177, <https://doi.org/10.1007/s40192-019-00132-9>.
- [34] P. Moazzen, A. Shahriari, S. Shamsdini, P. Seraj, F. Forooghi, Y. Aghayar, S. Shakerin, M.R. Purdy, M. Mohammadi, Optimum corrosion performance using microstructure design and additive manufacturing process control, *npj Mater. Degrad.* 9 (2025) 10, <https://doi.org/10.1038/s41529-024-00548-5>.
- [35] L. Parry, I.A. Ashcroft, R.D. Wildman, Understanding the effect of laser scan strategy on residual stress in selective laser melting through thermo-mechanical simulation, *Addit. Manuf.* 12 (2016) 1–15, <https://doi.org/10.1016/j.addma.2016.05.014>.
- [36] H. Azizi, A. Ebrahimi, N. Ofori-Opoku, M. Greenwood, N. Provatas, M. Mohammadi, Characterizing the microstructural effect of build direction during solidification of laser-powder bed fusion of Al-Si alloys in the dilute limit: a phase-field study, *Acta Mater.* 214 (2021) 116983, <https://doi.org/10.1016/j.actamat.2021.116983>.
- [37] J.R. Zhuang, Y.T. Lee, W.H. Hsieh, A.S. Yang, Determination of melt pool dimensions using DOE-FEM and RSM with process window during SLM of Ti6Al4V powder, *Opt. Laser Technol.* 103 (2018) 59–76, <https://doi.org/10.1016/j.optlastec.2018.01.013>.
- [38] S.I. Shahabadi, Z. Zhang, A. Keshavarzkermani, U. Ali, Y. Mahmoodkhani, R. Esmaeilzadeh, A. Bonakdar, E. Toyserkani, Heat source model calibration for

- thermal analysis of laser powder-bed fusion, *Int. J. Adv. Manuf. Technol.* 106 (2020) 3367–3379, <https://doi.org/10.1007/s00170-019-04908-3>.
- [39] L.E. Criales, Y.M. Arisoy, T. Özel, Sensitivity analysis of material and process parameters in finite element modeling of selective laser melting of Inconel 625, *Int. J. Adv. Manuf. Technol.* 86 (2016) 2653–2666, <https://doi.org/10.1007/s00170-015-8329-y>.
- [40] S.S. Sih, J.W. Barlow, THE PREDICTION OF THE THERMAL CONDUCTIVITY OF POWDERS, n.d.
- [41] S. Kazemi, Soleimani.M, Melting pool simulation of 316L samples manufactured by Selective Laser Melting method, comparison with experimental results, *International Journal of Thermal Science* 176 (2022).
- [42] D. Augusto De Moraes, A. Czekanski, THERMAL MODELING OF 304L STAINLESS STEEL FOR SELECTIVE LASER MELTING: LASER POWER INPUT EVALUATION, 2017. <http://asmedigitalcollection.asme.org/IMECE/proceedings-pdf/IMECE2017/58356/V002T02A033/2499021/v002t02a033-imece2017-72224.pdf>.
- [43] G. Zhu, A. Zhang, D. Li, Y. Tang, Z. Tong, Q. Lu, Numerical simulation of thermal behavior during laser direct metal deposition, *Int. J. Adv. Manuf. Technol.* 55 (2011) 945–954, <https://doi.org/10.1007/s00170-010-3142-0>.
- [44] S. Liu, H. Zhu, G. Peng, J. Yin, X. Zeng, Microstructure prediction of selective laser melting AISi10Mg using finite element analysis, *Mater. Des.* 142 (2018) 319–328, <https://doi.org/10.1016/j.matdes.2018.01.022>.
- [45] W. Zhang, G.G. Roy, J.W. Elmer, T. DebRoy, Modeling of heat transfer and fluid flow during gas tungsten arc spot welding of low carbon steel, *J. Appl. Phys.* 93 (2003) 3022–3033, <https://doi.org/10.1063/1.1540744>.
- [46] Z. Zhang, Y. Huang, A. Rani Kasinathan, S. Imani Shahabad, U. Ali, Y. Mahmoodkhani, E. Toyserkani, 3-Dimensional heat transfer modeling for laser powder-bed fusion additive manufacturing with volumetric heat sources based on varied thermal conductivity and absorptivity, *Opt Laser Technol* 109 (2019) 297–312. doi:10.1016/j.optlastec.2018.08.012.
- [47] M. Alimardani, E. Toyserkani, J.P. Huissoon, A 3D dynamic numerical approach for temperature and thermal stress distributions in multilayer laser solid freeform fabrication process, *Opt. Lasers Eng.* 45 (2007) 1115–1130, <https://doi.org/10.1016/j.optlaseng.2007.06.010>.
- [48] Z. Sun, W. Guo, L. Li, Numerical modelling of heat transfer, mass transport and microstructure formation in a high deposition rate laser directed energy deposition process, *Addit. Manuf.* 33 (2020) 101175, <https://doi.org/10.1016/j.addma.2020.101175>.
- [49] Z. Kazemi, M. Soleimani, H. Rokhgireh, A. Nayeibi, Melting pool simulation of 316L samples manufactured by Selective Laser Melting method, comparison with experimental results, *Int. J. Therm. Sci.* 176 (2022) 107538, <https://doi.org/10.1016/j.ijthermalsci.2022.107538>.
- [50] X. Kan, Y. Yin, D. Yang, W. Li, J. Sun, Micro pool characteristics of 316L and the influence of sulfur during SLM, *Opt. Laser Technol.* 142 (2021) 107136, <https://doi.org/10.1016/j.optlastec.2021.107136>.
- [51] E.-L. Papazoglou, N. Karkalos, A. Markopoulos, A comprehensive study on thermal modeling of SLM process under conduction mode using FEM, *Int. J. Adv. Manuf. Technol.* 111 (2020) 1–17, <https://doi.org/10.1007/s00170-020-06294-7>.
- [52] H.M. Khan, M.H. Dirikolu, E. Koç, Z.C. Oter, Numerical investigation of heat current study across different platforms in SLM processed multi-layer AISi10Mg, *Optik (stuttgart)* 170 (2018) 82–89, <https://doi.org/10.1016/j.ijleo.2018.05.081>.
- [53] A. Khan, S.H.I. Jaffery, S.Z. Hussain, Z. Anwar, S. Dilawar, Numerical and experimental characterization of melt pool in laser powder bed fusion of SS316L, *Integr Mater Manuf Innov* 12 (2023) 210–230, <https://doi.org/10.1007/s40192-023-00302-w>.
- [54] P.N.Q. Juan, J. Valencia, *Thermophysical Properties, Casting*, ASM Handbook (2008).
- [55] X. Li, W. Tan, Numerical investigation of effects of nucleation mechanisms on grain structure in metal additive manufacturing, *Comput. Mater. Sci* 153 (2018) 159–169, <https://doi.org/10.1016/j.commatsci.2018.06.019>.
- [56] S. Patel, M. Vlasea, Melting modes in laser powder bed fusion, *Materialia (oxf)* 9 (2020) 100591, <https://doi.org/10.1016/j.mtla.2020.100591>.
- [57] X. Zhang, W. Chen, J. Ren, Y. Wang, G. Huang, H. Zhang, Effects of processing parameters on mode and stability of laser welding, in: *Proc.SPIE*, 1996: pp. 343–349. doi:10.1117/12.237746.
- [58] W.E. King, H.D. Barth, V.M. Castillo, G.F. Gallegos, J.W. Gibbs, D.E. Hahn, C. Kamath, A.M. Rubenchik, Observation of keyhole-mode laser melting in laser powder-bed fusion additive manufacturing, *J. Mater.Process. Technol.* 214 (2014) 2915–2925, <https://doi.org/10.1016/j.jmatprotec.2014.06.005>.
- [59] M. Asta, C. Beckermann, A. Karma, W. Kurz, R. Napolitano, M. Plapp, G. Purdy, M. Rappaz, R. Trivedi, Solidification microstructures and solid-state parallels: recent developments, future directions, *Acta Mater.* 57 (2009) 941–971, <https://doi.org/10.1016/j.actamat.2008.10.020>.
- [60] S. Kou, *Welding Metallurgy*, MRS Bull. 28 (2003) 674–675, <https://doi.org/10.1557/mrs2003.197>.
- [61] W.-I. Cho, S.-J. Na, C. Thomy, F. Vollertsen, Numerical simulation of molten pool dynamics in high power disk laser welding, *J. Mater. Process. Technol.* 212 (2012) 262–275, <https://doi.org/10.1016/j.jmatprotec.2011.09.011>.
- [62] C. Zhao, K. Fezzaa, R.W. Cunningham, H. Wen, F. De Carlo, L. Chen, A.D. Rollett, T. Sun, Real-time monitoring of laser powder bed fusion process using high-speed X-ray imaging and diffraction, *Sci. Rep.* 7 (2017), <https://doi.org/10.1038/s41598-017-03761-2>.
- [63] C.A. Moreno-Camacho, J.R. Montoya-Torres, A. Jaegler, N. Gondran, Sustainability metrics for real case applications of the supply chain network design problem, *A Systematic Literature Review* (2019).
- [64] J. Yang, J. Han, H. Yu, J. Yin, M. Gao, Z. Wang, X. Zeng, Role of molten pool mode on formability, microstructure and mechanical properties of selective laser melted Ti-6Al-4V alloy, *Mater. Des.* 110 (2016) 558–570, <https://doi.org/10.1016/j.matdes.2016.08.036>.
- [65] K.M. Bertsch, G. Meric de Bellefon, B. Kuehl, D.J. Thoma, Origin of dislocation structures in an additively manufactured austenitic stainless steel 316L, *Acta Mater.* 199 (2020) 19–33, <https://doi.org/10.1016/j.actamat.2020.07.063>.
- [66] Y. Yang, X. Li, M.M. Khonsari, Y. Zhu, H. Yang, On enhancing surface wear resistance via rotating grains during selective laser melting, *Addit. Manuf.* 36 (2020) 101583, <https://doi.org/10.1016/j.addma.2020.101583>.
- [67] W. Shao, B. He, C. Qiu, Z. Li, Effect of hatch spacing and laser remelting on the formation of unique crystallographic texture of IN718 superalloy fabricated via laser powder bed fusion, *Opt. Laser Technol.* 156 (2022) 108609, <https://doi.org/10.1016/j.optlastec.2022.108609>.